



COMSATS University Islamabad, Pakistan

GAR OPTICALS

**A project presented to
COMSATS University Islamabad, Sahiwal Campus**

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of the requirement for the degree of**

Bachelor of Science in Computer Science (2022-2026)

By

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Executive Summary

In the traditional optical retail industry, customers frequently face challenges such as the inconvenience of visiting physical stores, the inability to visualize frames accurately before purchase, and a lack of personalized guidance when selecting eyewear. Existing manual operations for managing product inventories, processing orders, and handling customer queries are often inefficient, leading to fragmented management and a limited shopping experience. There is a critical need for a modern, digital solution that bridges the gap between the physical retail experience and the convenience of online shopping.

To address these inefficiencies, **GAR Optical**s has been developed as a comprehensive, cross-platform mobile application. It is designed to revolutionize the way users browse, virtually try on, and purchase optical glasses and lenses. By transitioning from traditional, manual retail processes to an automated and intelligent digital storefront, the system provides a seamless shopping environment that empowers customers while offering store owners a robust, data-driven platform for business administration.

The core functionality of **GAR Optical**s centers on the integration of advanced technologies. The application features a real-time Virtual Try-On module utilizing Augmented Reality (ARCore/ARKit) to allow users to visualize frames on their faces with high accuracy. The system incorporates an AI-based recommendation engine that suggests frames tailored to individual facial features and user preferences. Furthermore, the platform includes an integrated chatbot for 24/7 customer support and a web-based admin dashboard that enables business owners to manage inventory, track orders, monitor sales analytics, and update promotional offers efficiently.

Built on a modular, scalable client-server architecture, the application utilizes the Flutter framework for a unified cross-platform mobile experience, while Firebase serves as the backend infrastructure. This ensures real-time data synchronization, secure user authentication, and efficient cloud storage.

Ultimately, GAR Opticals redefines the customer journey by seamlessly blending accessibility with innovative technology. The platform not only streamlines day-to-day business operations but also empowers individuals to make confident, personalized eyewear choices from the comfort of their homes.

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Ahmad Raza

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AR	Augmented Reality
BaaS	Backend as a Service
FCM	Firebase Cloud Messaging
FYP	Final Year Project
IDE	Integrated Development Environment
PC	Personal Computer
SDD	Software Design Description
SDLC	Software Development Life Cycle
SDK	Software Development Kit
SRS	Software Requirements Specification
UI	User Interface

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CHAPTER 1

1. Introduction

This chapter serves as the foundational opening of the report, providing a comprehensive overview of the GAR Optical's project. It delineates the project's scope, highlights its relevance to the Computer Science curriculum, establishes the background and motivation behind the development of this intelligent optical retail platform, and reviews existing literature to contextualize the work within current technological trends.

1.1 Brief

GAR Optical's is an innovative, cross-platform mobile application designed to modernize the traditional retail experience for optical products, including eyeglasses and contact lenses. The project addresses the historical limitations of physical optical stores by integrating cutting-edge technologies, specifically Augmented Reality (AR) and Artificial Intelligence (AI).

The application is built using the Flutter framework to ensure a consistent, high-performance experience across both Android and iOS platforms, while utilizing Google Firebase as a robust backend-as-a-service (BaaS) for real-time data management, user authentication, and secure cloud storage. Key features include a real-time virtual try-on system powered by ARCore/ARKit, an AI-driven product recommendation engine, an integrated chatbot for 24/7 customer support, and a web-based dashboard for efficient administrative management. This project represents a shift toward a digital-first retail model, aiming to enhance consumer confidence, streamline inventory management, and provide a seamless, personalized shopping journey.

1.2 Relevance to Course Modules

The GAR Optical's project is a culmination of the core competencies acquired during the Bachelor of Science in Computer Science program. The relevance of this project to various course modules is demonstrated as follows:

- **Software Engineering:** This project applies the Software Development Life Cycle (SDLC) models, requirement gathering, and system design principles learned in Software Engineering modules to manage project complexity.
- **Human-Computer Interaction (HCI):** The design of the user interface and the AR-based try-on experience directly utilizes principles of HCI, focusing on usability, accessibility, and user-centric design to minimize cognitive load.
- **Artificial Intelligence & Machine Learning:** The implementation of the product

- recommendation engine and the NLP-based chatbot (via Dialogflow) leverages concepts from AI courses to provide intelligent, data-driven user experiences.
- **Mobile Computing:** The use of the Flutter framework and Dart language is a direct application of Mobile Computing coursework, demonstrating cross-platform development, state management, and device hardware integration.
- **Database Management Systems (DBMS):** The integration of Firebase Firestore (a NoSQL database) demonstrates practical knowledge of data modeling, synchronization, and secure data storage as studied in database and distributed systems modules.

1.3 Project Background

The traditional optical retail sector has long been characterized by a "try-before-you-buy" dependency, necessitating physical visits to retail outlets. This model creates several pain points: limited inventory exposure for users, the inconvenience of travel, and a lack of personalized guidance when selecting eyewear based on facial structure.

The idea behind GAR Opticals originated from the need to eliminate these friction points. By leveraging modern smartphone capabilities specifically depth sensing cameras and high performance mobile processors we can now virtually replicate the physical experience of trying on glasses. Voice-over IP and intelligent chatbots further augment this by replacing manual customer support with automated, instantaneous interactions. This project bridges the gap between the physical and digital, transforming the act of buying spectacles from a time-consuming physical process into an efficient, interactive digital experience. The system doesn't just display products; it aids the user in decision-making through smart filtering and virtual visualization.

1.4 Literature Review

The shift from traditional brick-and-mortar retail to digital commerce has been a primary focus of recent technological research. The following themes characterize the current research and industry trends relevant to GAR Opticals:

- **Augmented Reality (AR) in E-Commerce:** Recent studies in e-commerce indicate that Augmented Reality significantly improves customer engagement and reduces return rates by allowing users to visualize products in a real-world context before purchase. Research on ARCore and ARKit frameworks highlights the evolution of face-mesh tracking, which allows for precise

overlaying of 3D assets on facial geometry, essential for the "virtual try-on" feature utilized in this project.

- **AI-Driven Personalization:** The application of Machine Learning (ML) in retail is extensively documented as a method to increase conversion rates. Personalized recommendation engines, which analyze user behavior, face shape metrics, and historical purchase data, have become the standard for modern e-commerce platforms. Industry benchmarks suggest that AI-driven suggestions increase the average order value by providing relevant alternatives that align with individual user preferences.
- **Chatbots and Conversational AI:** Literature regarding customer service automation points toward the high efficiency of Natural Language Processing (NLP) models, such as Google's Dialogflow. Research confirms that integrating chatbots leads to lower operational costs for businesses and higher customer satisfaction due to 24/7 availability.
- **Mobile-First Retail Architecture:** Cloud-based backend architectures, particularly "Backend-as-a-Service" (BaaS) platforms like Firebase, are currently the preferred model for mobile application development. Academic literature emphasizes the benefits of such platforms in handling authentication, real-time database synchronization, and scalable hosting, which are crucial for maintaining the performance benchmarks of a modern, data-intensive mobile application.

1.5 Analysis from Literature Review

While the literature establishes that AR and AI are transformative for e-commerce, the implementation within the optical sector specifically for a localized market reveals a significant gap. Existing solutions are often limited to high-end global brands with massive technical infrastructure, leaving local optical retailers and consumers in developing markets without access to these sophisticated tools.

In the context of GAR Optical, our project directly addresses this gap. Unlike static online catalogs that provide only 2D images, our implementation of ARCore/ARKit offers a dynamic, user-centered experience that specifically mitigates the "uncertainty of fit" the primary cause of product returns in the eyewear industry. Our analysis suggests that by applying these global technological standards to a specialized, local optical retail platform, we can significantly lower the barrier to entry for smaller optical stores to adopt digitalization.

Furthermore, while the literature highlights that AI recommendation engines are standard in generic e-

commerce, our context utilizes this technology to solve a specific optical problem: the correlation between facial metrics and frame geometry. We are not just recommending products based on purchase history; we are synthesizing facial geometry data (via the AR module) to suggest frames that are ergonomically and aesthetically suitable. This analytical approach positions GAR Optical as a more intelligent, targeted solution compared to the generic "collaborative filtering" models found in broader literature.

1.6 Methodology and Software Lifecycle for GAR Optical

To ensure the successful delivery of a complex system involving AI, AR, and real-time database synchronization, this project adopts an **Agile-Iterative Software Development Life Cycle (SDLC)**.

The complexity of integrating specialized SDKs such as ARCore for real-time tracking and Dialogflow for natural language processing requires a methodology that accommodates continuous feedback and incremental refinement. The development lifecycle for GAR Optical is organized into distinct, recurring phases:

- **Requirements Engineering & Planning:** Defining user stories, system requirements, and technical constraints.
- **Iterative Design & Prototyping:** Designing the modular system architecture and UI/UX wireframes (Figma).
- **Implementation (Sprints):** Developing the core application components (User Authentication, Product Catalog, AR Try-On, AI Chatbot) in iterative sprints.
- **Testing & Validation:** Conducting unit testing and integration testing at the end of every sprint to ensure feature stability.
- **Deployment & Evaluation:** Assessing the performance of features against user requirements and feedback before proceeding to the next iteration.

This model ensures that the project remains adaptable, allowing for the fine-tuning of the AR visualization accuracy and AI response times throughout the development process.

1.6.1 Rationale behind Selected Methodology

The selection of the **Agile-Iterative SDLC** and **Object-Oriented Methodology** was strategic, driven by the unique technical demands of the GAR Optical project:

- **Flexibility in Research-Heavy Development:** Projects integrating AR and AI are inherently unpredictable; they require a "try-and-adjust" approach rather than the rigid, sequential nature

of the Waterfall model. The iterative model allows us to test AR face-mapping accuracy in a lab environment and refine the logic before full-scale integration.

- **Modular Complexity Management:** By utilizing an Object-Oriented approach, the system is decomposed into independent entities (User, Product, ARAsset, Transaction). This modularity allows the development team to isolate bugs within specific modules (e.g., debugging the chatbot without impacting the payment gateway) and facilitates easier code maintenance and future scalability.
- **Stakeholder Alignment:** Agile philosophy emphasizes frequent demonstration of functional components. This ensures that the progress of the project is consistently aligned with academic supervisor expectations and functional requirements, preventing the accumulation of technical debt or deviation from the core project goals.
- **Efficiency in Resource Utilization:** Given the scope of this project, the iterative model enables the team to prioritize high-value features (like the Product Catalog and AR Try-On) in early sprints, ensuring that a functional MVP (Minimum Viable Product) is always available, even if advanced features like complex AI analytics are refined in later cycles.
- **Risk Mitigation through Early Integration:** Integrating specialized SDKs (such as ARCore for facial mapping and external payment APIs) presents significant technical risk. The Agile-Iterative approach facilitates the "Fail-Fast" principle, where technical hurdles—such as AR latency or payment gateway authentication errors—are identified and resolved in the early sprints. This prevents the project from reaching the final phase with unresolvable dependencies, a common pitfall of the traditional Waterfall model in student research projects.
- **Continuous Quality Assurance:** Unlike the traditional model where testing occurs only after the implementation phase, the iterative methodology ensures that each functional module (e.g., the product catalog or the chatbot) undergoes rigorous unit and integration testing before the next sprint commences. This creates a high level of code stability, ensuring that as the GAR Opticals system grows in complexity, new features do not destabilize existing, proven modules.
- **User-Centric UI/UX Refinement:** Developing an optical retail application requires high user trust, particularly concerning the accuracy of the AR virtual try-on.

CHAPTER 2

2. Problem Definition

This chapter identifies the core challenges inherent in the traditional optical retail sector. By analyzing the limitations of existing manual systems, we establish the necessity for the GAR Optical's digital solution. The chapter also delineates the tangible deliverables and the functional scope of the proposed system.

2.1 Problem Statement

The contemporary optical retail industry, while essential for vision health, remains largely tethered to traditional, manual business models. Consumers today are accustomed to the convenience of e-commerce, yet the process of purchasing prescription eyewear remains fraught with inefficiencies.

Primary problems identified include:

- **Visualization Uncertainty:** Users lack the ability to accurately judge how a frame fits their facial structure without physically visiting a store, leading to low customer confidence and high return rates.
- **Fragmented Information:** Customers often struggle to access information regarding lens types, pricing, and availability, relying entirely on physical store availability.
- **Operational Inefficiency:** Retailers relying on manual record-keeping struggle with real-time inventory management, data reconciliation, and providing consistent, 24/7 customer support.
- **Lack of Personalization:** Existing systems fail to leverage data-driven insights to provide frame recommendations based on unique user facial metrics

2.2 Deliverables and Development Requirements

The development of GAR Optical's is aimed at replacing the aforementioned manual processes with a unified digital platform. The project deliverables are categorized into software applications and technical documentation.

Table 2.1 Project Deliverables

Category	Deliverable Item	Description
Software	Customer Mobile Application	Android/iOS app with AR try-on and chatbot.
Software	Admin Web Dashboard	Management portal for inventory, orders, and sales.
Backend	Firebase Infrastructure	Firestore database, Auth, and Storage.
Documentation	SRS Document	Detailed functional requirements.
Documentation	SDD Document	Architectural and design diagrams.

Development Requirements:

- **Hardware:** Development workstations, testing smartphones (Android/iOS) with depth-sensing capabilities.
- **Software Tools:** Flutter SDK (Dart), Android Studio, Firebase CLI, Figma (for UI/UX design), Git/GitHub (for version control).

2.3 Current System

The current system utilized by most small-to-medium optical shops is a purely manual, offline-first process. In this traditional workflow, the customer must physically visit the store, manually try on frames, and consult with the salesperson for advice. The inventory management is often done via physical ledgers or disjointed spreadsheet software, which provides no real-time data sync. The current manual system relies on physical ledgers and offline workflows, leading to significant delays and data inaccuracies. Customers must physically visit the store to browse inventory, which often results in "decision fatigue" and lost sales opportunities. Administratively, the lack of real-time synchronization makes tracking stock and customer prescriptions difficult and error-prone. Furthermore, without a digital platform, there is no way to engage customers once they leave the store, creating a major gap in service and business growth.

The process flow of the current manual system is depicted in Figure 2.1 below.

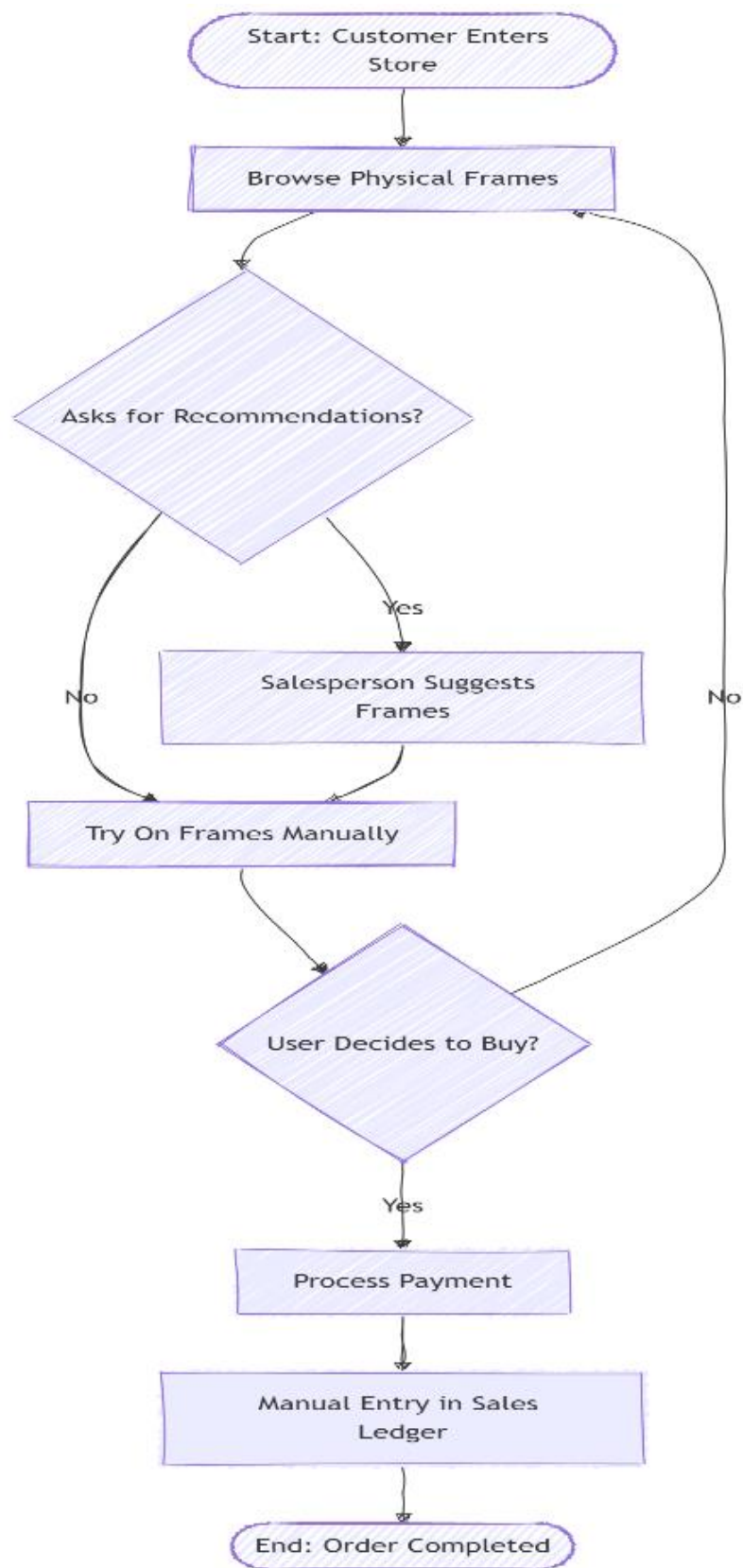


Figure 2:1 Workflow of the Current Manual System

Key Deficiencies of the Current System:

- **Limited Reach:** The system is constrained by the store's physical location and working hours.
- **Error-Prone Data Entry:** Manual record-keeping is susceptible to human error, leading to stock discrepancies.
- **Lack of Analytics:** Store owners have no digital way to track which products are trending or to perform predictive restocking.
- **No Customer Engagement:** Once the customer leaves the store, there is no digital channel for them to track orders or receive personalized support, resulting in low customer retention.
- **Inefficient Product Visualization:** Traditional retail limits customers to the inventory currently on display. Trying on multiple physical frames is time-consuming and raises hygiene concerns, often leading to "decision fatigue" before the customer can find the perfect match. There is currently no mechanism to instantly visualize how different frame styles suit the user's facial structure.
- **Data Vulnerability and Poor Accessibility:** Physical ledger books and local, disconnected files are vulnerable to loss, damage, or unauthorized access. Furthermore, this data is siloed, making it impossible for administrators to access vital business records such as daily sales or inventory levels remotely or in real-time.
- **High Administrative Overhead:** Staff members are forced to spend significant time on repetitive administrative tasks, such as answering basic product queries or manually documenting sales, rather than focusing on high-value tasks like complex prescription consultations or personalized customer service.
- **Scalability Bottlenecks:** The current manual processes create a "ceiling" for growth. As the volume of customers or product SKUs increases, the current system becomes unable to cope, leading to increased delays and a degradation of the service quality that the business aims to provide.

These cumulative deficiencies create significant friction in both the administrative operations and the customer shopping experience. The lack of digital integration not only hampers the store's ability to compete in a modern, tech-driven retail environment but also prevents the business from leveraging data-driven insights to maximize profitability. Consequently, there is an urgent need for the proposed GAR OPTICALS system a solution designed to replace these fragmented, manual processes with a centralized, automated, and AR-enhanced platform.

CHAPTER 3

3. Requirement Analysis

This chapter serves as a comprehensive bridge between the identified problems and the proposed solution's technical design. It systematically breaks down the requirements of GAR Optical's into use cases, functional specifications, and non-functional constraints.

3.1 Use Case Diagram

The Use Case Diagram defines the behavioral requirements of the system by modeling the interaction between the system's functions and its external users (actors). For GAR Optical's, we have defined two primary actors: the Customer, who seeks an enhanced digital shopping experience, and the Administrator, who manages business operations.

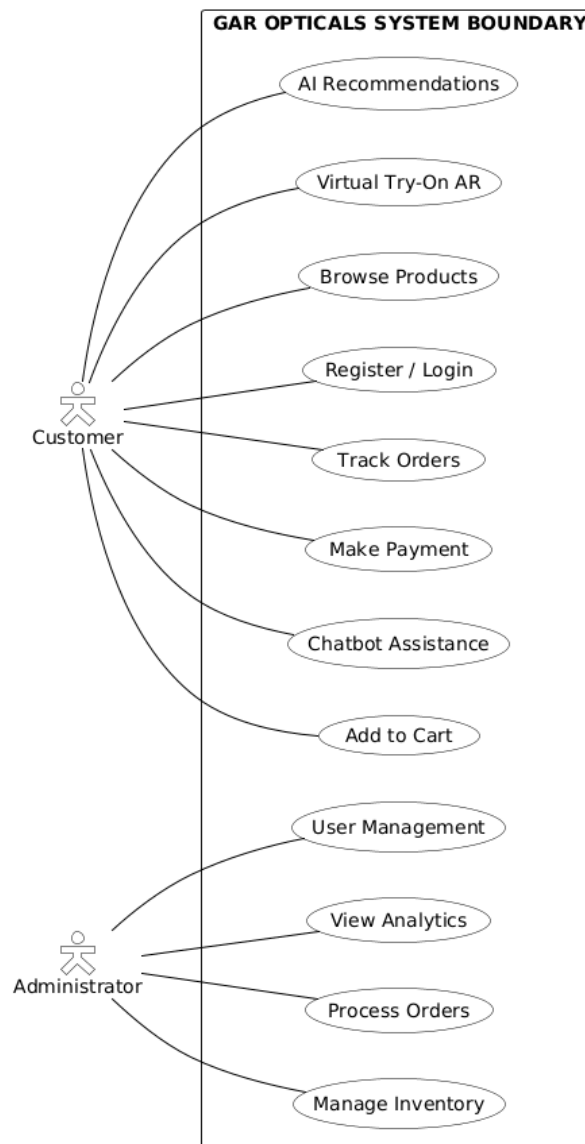


Figure 3:1 Use Case Diagram for GAR Optical's

The Use Case Diagram for GAR OPTICALS illustrates how two primary actors—Customer and Administrator interact with the system to perform optical retail operations. The diagram encloses all features within the system boundary labeled GAR OPTICALS, showing which actions are provided by the application and which are initiated by external users.

Customer interacts with the application to complete shopping tasks: account registration/login, browsing and filtering products, virtual try-on (AR), AI-based recommendations, adding items to wishlist/cart, secure payment, order tracking, product reviews, and chatbot assistance for real-time support. These use cases ensure an engaging, personalized, and self-service experience for purchasing glasses and lenses.

Administrator accesses the web-based dashboard to manage the business side of the system. Admin use cases include managing products (add/edit/delete), managing users, processing orders, creating promotions/offers, and viewing reports & analytics. These actions maintain catalog accuracy, ensure smooth order fulfillment, and provide data-driven insights for decision-making.

Several relationships connect the use cases for clarity and reuse. For example, Secure Payment and Order Tracking are included by Checkout; Notifications (via FCM) extend Order Processing and Promotions to inform users in real time; AI-Based Recommendations and Virtual Try-On (AR) support Product Viewing to improve selection quality. Together, the diagram summarizes the end-to-end flow from product discovery to purchase and post-purchase management within GAR OPTICALS.

3.2 Use Case Descriptions

3.2.1 Use Case 1: User Management

1. **Use Case ID:** UC-1
2. **Use Case Name:** User Management
3. **Actors:** * Primary Actor: Administrator
 - Secondary Actor: System
4. **Description:** The Administrator manages all user accounts, including registration, updates, and deletion of customer profiles. The system validates all user inputs, ensures authentication through Firebase, and maintains access control for secure data management.

5. **Trigger:** Admin selects “User Management” from the dashboard to create, update, or remove customer accounts.
6. **Preconditions:** Admin must be logged in with valid credentials.
7. **Postconditions:** User records are successfully added, modified, or deleted in the database, and logs are updated in Firebase.
8. **Normal Flow:**
 - Admin navigates to the User Management module.
 - Admin selects Add/Edit/Delete option.
 - System validates inputs.
 - System performs the selected action.
 - Confirmation message displayed.
9. **Alternative Flows:**
 - If the email already exists, the system prompts to update the record.
 - If fields are missing, the system requests completion.
10. **Exceptions:** E1: If Firebase connection fails, the operation is canceled.
11. **Business Rules:**
 - BR-1: Only Admin can manage user accounts.
 - BR-2: All user operations must be logged in the system database.

3.2.2 Use Case 2: Product Management

1. **Use Case ID:** UC-2
2. **Use Case Name:** Product Management
3. **Actors:** * Primary Actor: Admin
 - Secondary Actor: System
4. **Description:** Admin manages the eyewear product catalog, including adding, editing, or deleting items. The system validates product details, updates stock, and synchronizes all changes in Firebase Firestore.

5. **Trigger:** Admin selects “Manage Products” from the dashboard.
6. **Preconditions:** Admin is logged in and authorized.
7. **Postconditions:** Product catalog is updated and visible to all customers.
8. **Normal Flow:**
 - Admin clicks “Add Product.”
 - Admin fills in name, price, and image details.
 - System validates and saves the product in Firestore.
 - Admin confirms successful entry.
9. **Alternative Flows:**
 - If image upload fails, the system prompts for re-upload.
 - If duplicate name exists, the system suggests editing the existing product.
10. **Exceptions:** E1: Network loss interrupts product update.
11. **Business Rules:**
 - BR-1: Only Admin can modify product data.
 - BR-2: Product images must be under 5MB and in JPEG/PNG format.

3.2.3 Use Case 3: Virtual Try-On (AR)

1. **Use Case ID:** UC-3
2. **Use Case Name:** Virtual Try-On (AR)
3. **Actors:** * Primary Actor: Customer
 - Secondary Actor: System
4. **Description:** The Customer can use the AR feature to virtually try on glasses before purchase. The camera opens, overlays selected frames, and allows preview from multiple angles using ARCore.
5. **Trigger:** Customer clicks “Try On” under a product listing.
6. **Preconditions:** Device must support ARCore and have a functional camera.
7. **Postconditions:** The selected frame is displayed virtually on the user’s face.

8. Normal Flow:

- Customer selects “Try On.”
- Camera launches via ARCore.
- System overlays the selected frame.
- Customer views and adjusts fit.

9. Alternative Flows:

- If AR fails to load, the system shows a static image preview.

10. Exceptions: E1: ARCore is not supported on the device.

11. Business Rules:

- BR-1: AR feature is available only on compatible devices.
- BR-2: User must grant camera access to use the AR module.

3.2.4 Use Case 4: Order and Payment Management

1. Use Case ID: UC-4

2. Use Case Name: Order and Payment Management

3. Actors: * Primary Actor: Customer

- Secondary Actor: System

4. Description: Customers can place orders, make secure online payments, and track order status. The system verifies transaction details through integrated payment APIs and stores order information in Firebase.

5. Trigger: Customer clicks “Proceed to Checkout.”

6. Preconditions: User must be logged in and have selected items in the cart.

7. Postconditions: Payment confirmation and order tracking ID are generated.

8. Normal Flow:

- Customer reviews cart.
- Customer selects payment method (Credit/Debit Card, Wallet).
- System processes payment securely.

- Confirmation and invoice displayed.

9. Alternative Flows:

- If payment fails, retry or select another method.

10. Exceptions: E1: Payment gateway timeout or failure.

11. Business Rules:

- BR-1: All payments must use verified gateways (Stripe/Razorpay).
- BR-2: Refunds must follow admin approval.

3.2.5 Use Case 5: Chatbot and Customer Support

1. Use Case ID: UC-5

2. Use Case Name: Chatbot and Customer Support

3. Actors: * Primary Actor: Customer

- Secondary Actor: Chatbot System

4. Description: Customers can interact with the built-in chatbot for queries, product information, or assistance with orders. The system uses Dialogflow to provide real-time responses and log interactions for analytics.

5. Trigger: Customer clicks “Chat with Support.”

6. Preconditions: User must have internet access.

7. Postconditions: Customer receives appropriate responses or escalations to the admin.

8. Normal Flow:

- Customer opens the chatbot.
- Customer enters a question or issue.
- Chatbot responds or redirects to admin if unresolved.

9. Alternative Flows:

- If the Dialogflow service is unavailable, the system displays a fallback message.

10. Exceptions: E1: Connection error interrupts the chat session.

11. Business Rules:

- BR-1: Chatbot must respond within 3 seconds.
- BR-2: Admin must review logs weekly for unresolved queries.

3.3 Functional Requirements

Table 3.1 Summary of Functional Requirements

Requirement ID	Title	Priority
FR-1.1	User Account Management	High
FR-1.2	User Authentication	High
FR-1.3	User Data Validation	Medium
FR-2.1	Product Catalog Management	High
FR-2.2	Product Filtering and Search	High
FR-2.3	Product Display and Details	Medium
FR-3.1	AR Try-On Activation	High
FR-3.2	Frame Fit Adjustment	Medium
FR-3.3	AR Failure Handling	Medium
FR-4.1	Order Placement	High
FR-4.2	Secure Payment Integration	High
FR-4.3	Order Tracking and Notifications	Medium
FR-5.1	Chatbot Interaction	High

Requirement ID	Title	Priority
FR-5.2	Chat History and Logging	Medium
FR-5.3	Fallback and Escalation	High

3.3.1 User Management

Table 3.2 User Account Management (FR-1.1)

Field	Description
Requirement ID	FR-1.1
Title	User Account Management
Requirement	The admin shall be able to create, update, and delete user accounts for customers through the dashboard.
Source	System Design / Admin Role
Rationale	To manage customer access and maintain data consistency in Firebase Authentication.
Business Rule	Only authorized admins can manage accounts; each account must have a unique email ID.
Dependencies	Firebase Authentication, Firestore Database
Priority	High

Table 3.3 User Authentication (FR-1.2)

Field	Description
Requirement ID	FR-1.2
Title	User Authentication
Requirement	The system shall validate user credentials during login and provide password reset functionality through Firebase.
Source	Authentication Module
Rationale	Ensures secure access control and account recovery.
Business Rule	Passwords must meet security standards (min. 8 characters, 1 uppercase, 1 digit).
Dependencies	Firebase Authentication
Priority	High

Table 3.4 User Data Validation (FR-1.3)

Field	Description
Requirement ID	FR-1.3
Title	User Data Validation
Requirement	The system shall prevent duplicate account creation and validate all input fields before storing user data.
Source	Admin & User Module
Rationale	Maintains data accuracy and prevents redundancy in the database.

Field	Description
Business Rule	All input fields must be verified for completeness before submission.
Dependencies	Firestore Database
Priority	Medium

3.3.2 Product Management

Table 3.5 Product Catalog Management (FR-2.1)

Field	Description
Requirement ID	FR-2.1
Title	Product Catalog Management
Requirement	The admin shall be able to add, edit, and delete eyewear products including name, price, image, and description.
Source	Admin Dashboard
Rationale	Ensures the product catalog remains accurate and up to date.
Business Rule	All product updates must synchronize with Firestore in real-time.
Dependencies	Firestore Database, Cloud Storage
Priority	High

Table 3.6 Product Filtering and Search (FR-2.2)

Field	Description
Requirement ID	FR-2.2
Title	Product Filtering and Search
Requirement	Users shall be able to search products and apply filters based on price, frame type, or brand.
Source	User Application
Rationale	Improves navigation and user shopping experience.
Business Rule	Search results should update dynamically with applied filters.
Dependencies	Firestore Query System
Priority	High

Table 3.7 Product Display and Details (FR-2.3)

Field	Description
Requirement ID	FR-2.3
Title	Product Display and Details
Requirement	The system shall display each product's image, specifications, and price details dynamically fetched from Firestore.
Source	Frontend Design
Rationale	Ensures customers can make informed purchasing decisions.

Field	Description
Business Rule	Product details must load within 2 seconds for a smooth experience.
Dependencies	Firebase Firestore, Flutter Frontend
Priority	Medium

3.3.3 Virtual Try-On (AR Module)

Table 3.8 AR Try-On Activation (FR-3.1)

Field	Description
Requirement ID	FR-3.1
Title	AR Try-On Activation
Requirement	The system shall activate the device camera and overlay the selected glasses frame using ARCore.
Source	AR Module
Rationale	Provides a real-time preview of eyewear on the user's face.
Business Rule	Feature is accessible only on AR-compatible devices.
Dependencies	ARCore SDK, Flutter Plugin
Priority	High

Table 3.9 Frame Fit Adjustment (FR-3.2)

Field	Description
Requirement ID	FR-3.2
Title	Frame Fit Adjustment
Requirement	The user shall be able to adjust the glasses' position and scale during virtual try-on.
Source	User Interface
Rationale	Improves accuracy of visual fit and enhances user satisfaction.
Business Rule	System must maintain frame alignment according to face tracking data.
Dependencies	ARCore Face Mesh, Camera Access
Priority	Medium

Table 3.10 AR Failure Handling (FR-3.3)

Field	Description
Requirement ID	FR-3.3
Title	AR Failure Handling
Requirement	If AR mode fails to load, the system shall display a static product preview instead.
Source	Error Handling Module
Rationale	Ensures usability even on unsupported devices.

Field	Description
Business Rule	Fallback preview must show multiple angle images of the frame.
Dependencies	Product Catalog, ARCore
Priority	Medium

3.3.4 Order and Payment Management

Table 3.11 Order Placement (FR-4.1)

Field	Description
Requirement ID	FR-4.1
Title	Order Placement
Requirement	The system shall allow users to place orders for selected products after confirming address and payment details.
Source	Checkout Module
Rationale	Facilitates smooth order processing and confirmation.
Business Rule	Order ID must be generated automatically upon payment success.
Dependencies	Payment Gateway API, Firestore
Priority	High

Table 3.12 Secure Payment Integration (FR-4.2)

Field	Description
Requirement ID	FR-4.2
Title	Secure Payment Integration
Requirement	The system shall process payments using verified gateways such as Stripe or Razorpay.
Source	Payment Gateway
Rationale	Ensures secure and reliable transaction processing.
Business Rule	Transaction data must be encrypted using SSL.
Dependencies	Stripe/Razorpay API, Firebase Functions
Priority	High

Table 3.13 Order Tracking and Notifications (FR-4.3)

Field	Description
Requirement ID	FR-4.3
Title	Order Tracking and Notifications
Requirement	Users shall be able to track their order status and receive notifications for dispatch and delivery.
Source	Tracking Module
Rationale	Keeps users informed and increases post-purchase engagement.

Field	Description
Business Rule	Notifications must trigger via Firebase Cloud Messaging (FCM).
Dependencies	Firebase FCM, Firestore
Priority	Medium

3.3.5 Chatbot and Customer Support

Table 3.14 Chatbot Interaction (FR-5.1)

Field	Description
Requirement ID	FR-5.1
Title	Chatbot Interaction
Requirement	The system shall enable real-time chatbot communication for customer support using Dialogflow.
Source	Chatbot Module
Rationale	Provides instant query resolution for customers.
Business Rule	Chatbot must reply within 3 seconds of query submission.
Dependencies	Dialogflow API, Internet Connectivity
Priority	High

Table 3.15 Chat History and Logging (FR-5.2)

Field	Description
Requirement ID	FR-5.2
Title	Chat History and Logging
Requirement	The system shall log all chatbot conversations for admin review and performance tracking.
Source	Support Module
Rationale	Helps improve response accuracy and service quality.
Business Rule	Chat logs must be stored securely for at least 30 days.
Dependencies	Firebase Storage, Dialogflow
Priority	Medium

Table 3.16 Fallback and Escalation (FR-5.3)

Field	Description
Requirement ID	FR-5.3
Title	Fallback and Escalation
Requirement	If the chatbot cannot resolve a query, it shall escalate the request to the admin via a support ticket.
Source	Customer Support
Rationale	Ensures no customer issue remains unresolved.

Field	Description
Business Rule	Unanswered queries must be flagged for manual review within 24 hours.
Dependencies	Chatbot Module, Admin Dashboard
Priority	High

3.4 Non-Functional Requirements

Non-functional requirements define the quality attributes, performance expectations, and operational standards of the GAR OPTICALS system. These requirements ensure that the system not only performs its intended functions but also provides efficiency, reliability, and user satisfaction.

3.4.1 Usability

- The GAR OPTICALS application is designed with user-friendliness and accessibility as core priorities. The interface will be simple, intuitive, and visually appealing to ensure that users of all age groups can navigate the system effortlessly. The Flutter framework provides a responsive and uniform interface across Android and iOS platforms.
- The app uses a clean layout with categorized product listings, easy search and filter options, and a smooth checkout process. Icons and menus are clearly labeled, ensuring that even first-time users can complete tasks without prior guidance. Moreover, the AR-based try-on feature enhances interactivity and makes the shopping experience more engaging and personalized.
- Usability testing will be conducted to gather feedback on layout, flow, and ease of navigation, ensuring continuous improvement and satisfaction for both customers and administrators.

3.4.2 Performance

- The performance of GAR OPTICALS will be optimized to deliver smooth and responsive interactions. The system will maintain a response time of under 2 seconds for most user operations such as product browsing, adding items to cart, or fetching

recommendations.

- The backend, powered by Firebase, ensures high availability and efficient data retrieval using indexed queries and optimized cloud functions. For real-time operations like AR rendering and chatbot responses, the system leverages asynchronous processing and lazy loading to minimize delays.
- The application will be capable of supporting up to 500 concurrent users without noticeable performance degradation. Resource usage (RAM, CPU, and network) will be monitored to maintain a balance between responsiveness and energy efficiency on mobile devices.

3.4.3 Scalability

- The system architecture is designed for scalability to accommodate increasing users, products, and transactions. Since Firebase provides a cloud-based infrastructure, it allows the database, storage, and hosting services to scale automatically based on user demand.
- As the number of optical products and customer interactions grows, the database structure can easily expand without requiring manual reconfiguration. The modular design of GAR OPTICALS also supports future integration with external APIs such as courier services, additional payment gateways, or AI analytics tools.
- Scalability will ensure that the system remains stable and efficient even as traffic, data volume, and functionality increase over time.

3.4.4 Availability

- The GAR OPTICALS system will maintain an uptime of 99.5%, ensuring that users can access the application and its services at almost all times. Firebase Hosting and Firestore databases guarantee redundancy and automatic failover in case of regional outages.
- In the event of a network issue, the app will offer read-only offline access to previously cached product data. Administrators can perform non-critical operations even during maintenance periods.
- Scheduled maintenance windows and updates will be managed to minimize downtime, ensuring consistent access and customer trust.

3.4.5 Compatibility

- The system will be cross-platform compatible, functioning seamlessly on both Android

and iOS devices. Built on Flutter, it ensures consistent UI/UX design, layout responsiveness, and behavior across different operating systems and screen resolutions.

- The admin dashboard will run on all major browsers, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari. Compatibility testing will verify responsiveness on desktops, tablets, and smartphones.
- In terms of integration, GAR OPTICALS will maintain compatibility with essential APIs, including Firebase, Dialogflow, ARCore, and Stripe/Razorpay, ensuring uninterrupted functionality across system updates.

3.4.6 Maintainability

- The application is structured with maintainability in mind, allowing future developers to easily update or enhance the system. The codebase follows modular design principles and adheres to Flutter best practices, ensuring that each module (user management, product catalog, AR module, chatbot, payment system) operates independently.
- Version control through GitHub enables collaborative development and proper tracking of changes. Firebase's centralized data management further simplifies maintenance by reducing dependencies on physical servers.
- The system documentation will include setup guides, API references, and workflow diagrams to help future maintainers make efficient updates or bug fixes without disrupting other modules.

CHAPTER 4

4. Design and Architecture

This chapter presents the architectural blueprint of the GAR OPTICALS system. It details the interaction between client and server, the data structure, and the process workflows that govern the platform.

4.1 System Architecture

The system's architecture is organized to enable seamless, real-time interactions between the client (users) and the cloud-based server, leveraging AR and AI for core features.

4.1.1 Frontend

- **UI/UX:** Provides the user interface (UI) for browsing the product catalog, user profile management, and interacting with the AR try-on and chatbot features.
- **Framework:** Developed using Flutter, it handles local processing and presentation, ensuring a consistent and responsive experience on both Android and iOS devices.
- **Navigation:** Includes search and filter functions to help buyers find products based on criteria such as frame style, price, and color.

4.1.2 Backend

- **Technology:** Primarily Firebase Backend as a Service (BaaS).
- **Function:** Manages secure User Authentication, handles Order Tracking, coordinates specialized API calls (e.g., Payment Gateway, Chatbot, AI), and manages Push Notifications via FCM.
- **Key Aspect:** The Firebase backend provides real-time data synchronization and scalability without the need for managing custom server infrastructure.

4.1.3 Database

- **Technology:** Firebase Firestore (NoSQL) and Firebase Storage.
- **Function:** Firestore stores essential structured data, including user accounts, product listings, order records, and reviews. Firebase Storage is used to host large assets, such as product images, 3D AR models, and other media files.

4.1.4 AR/AI Modules

- **AR (Virtual Try-On):** Uses the device camera and relevant frameworks (ARCore/ARKit) to detect facial geometry and accurately overlay the selected frame

onto the user's face in real time.

- **AI (Recommendation System):** Employs an AI algorithm (potentially leveraging TensorFlow Lite or MediaPipe) to analyze user preferences and facial features, providing detailed, personalized frame suggestions.

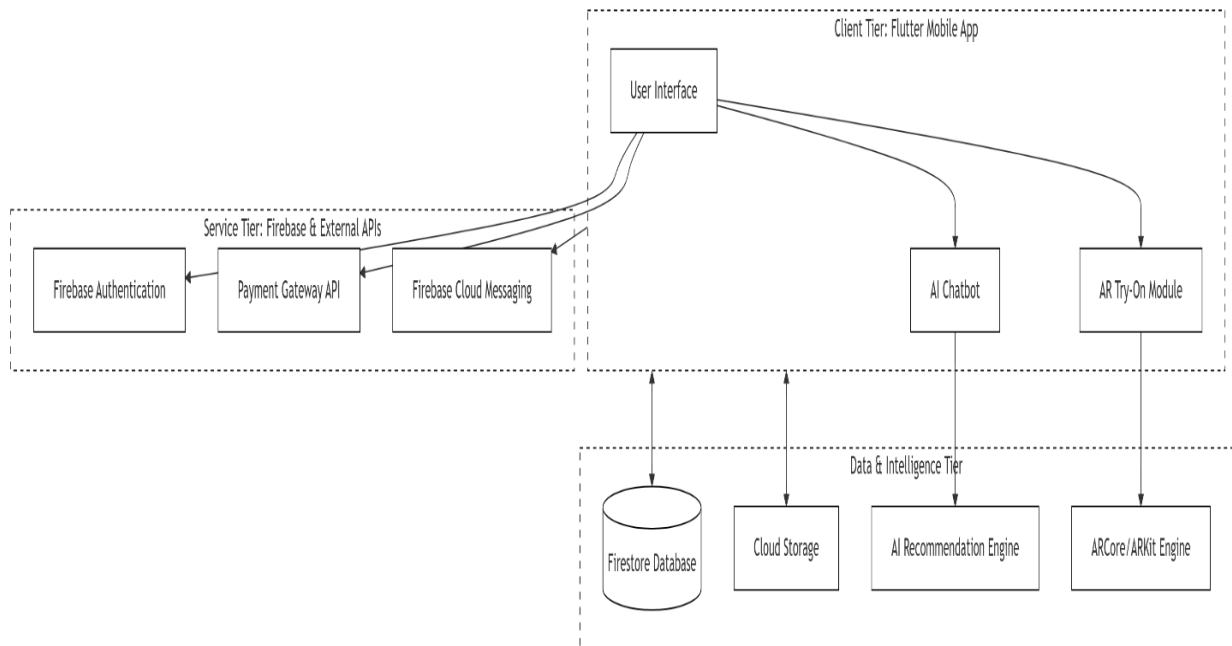


Figure 4:1 Architecture Design

The diagram above illustrates the flow of interaction, showing how the mobile application's UI connects to the Firebase Backend, which coordinates the core modules for AR, AI, Payment, and Database management.

4.2 Process Flow

The process flow for GAR OPTICALS is structured to provide smooth, interactive experiences for product visualization, purchasing, and order management.

4.2.1 Virtual Try-On and Recommendation Flow

This flow outlines the steps for a user to visualize products and receive personalized suggestions:

- **Product Selection:** User browses the product catalog and selects a frame to virtually try on.
- **AR Initialization:** The system activates the mobile camera and uses AR frameworks (ARCore/ARKit) to capture the user's face and map facial features.
- **Real-time Overlay:** The selected frame is accurately overlaid onto the user's face, allowing the user to view it from different angles.

- **AI Recommendation:** Concurrently, the AI model processes the facial data and user profile/browsing history to generate the most suitable frame suggestions.
- **Output Display:** The system displays the AR visualization along with the list of personalized recommendations.

4.2.2 Purchasing and Order Flow

This flow describes the steps for a user to complete a secure transaction and track their purchase:

- **Add to Cart:** The user selects a product (frame or lens) and proceeds to the secure checkout.
- **Payment Processing:** The system connects to the integrated Payment Gateway (e.g., Razorpay/Stripe) to securely validate and process the transaction details.
- **Transaction Recording:** The Secure Transaction Algorithm logs all order details, buyer/seller IDs, and the payment status in the Firebase Database.
- **Status Update:** The listing status is updated, and the user receives a confirmation notification via Firebase Cloud Messaging (FCM).
- **Order Tracking:** The user can view the real-time status of their order (e.g., Pending, Packed, Shipped, Delivered) through the app's order tracking feature.

4.2.3 Exception Handling and Security Protocols

The purchasing flow is designed with robust error-handling mechanisms to ensure data consistency and financial security in the event of unforeseen system interruptions:

- **Payment Failure Recovery:** If the communication between the application and the Payment Gateway (Stripe/Razorpay) is interrupted, the system is designed to trigger a "retry" prompt rather than creating a partial order record. This prevents the generation of "ghost orders" in the database.
- **Data Consistency and Atomicity:** During the checkout process, the system employs atomic write operations in Firestore. This ensures that the product inventory count and the customer's order record are updated simultaneously; if one operation fails, the entire transaction is rolled back to prevent stock discrepancies.
- **Security of Transactional Data:** To comply with standard payment security practices, sensitive user information (such as card details) is never stored directly on the application's local storage. Instead, the application utilizes secure tokenization

provided by the payment gateway, and all data transmission between the mobile client and the backend is encrypted using industry-standard SSL/TLS protocols.

- **Offline Order Queueing:** In scenarios where the user experiences a network drop immediately after initiating a payment, the application maintains a local cache of the order attempt. Upon reconnection, the system verifies the transaction status with the server before finalizing the order, ensuring that no customer payment is lost due to intermittent connectivity
- **Role-Based Access Control (RBAC):** To prevent unauthorized data manipulation, the system implements strict role-based access controls within the Firebase environment. Administrative functions, such as inventory editing and user management, are protected by custom claims that verify the user's identity before granting access to sensitive dashboard modules.
- **Secure API Communication:** All communications between the Flutter frontend and the Firebase backend are signed and validated using JSON Web Tokens (JWT). This ensures that even if a request is intercepted, it cannot be replayed or modified, protecting the system from man-in-the-middle (MITM) attacks and ensuring that only the official GAR OPTICALS application can interact with the server.

4.2.4 Audit Trails and Compliance

To maintain operational transparency and adhere to modern data protection regulations, the system generates comprehensive audit trails for all critical purchasing and exception events. Any administrative modification, operational failure, or unexpected termination during the checkout phase is securely logged in the database. This infrastructure ensures that administrators can quickly audit transaction anomalies, investigate system faults, and execute targeted diagnostic recoveries without compromising user privacy. Furthermore, all audit records are encrypted at rest within the cloud infrastructure, preventing any unauthorized tampering with transaction logs or user-specific data. The system also runs automated weekly security evaluations to identify potential vulnerabilities. This compliance-first approach guarantees that the platform meets strict industry benchmarks for consumer data protection, transparency, and operational integrity.

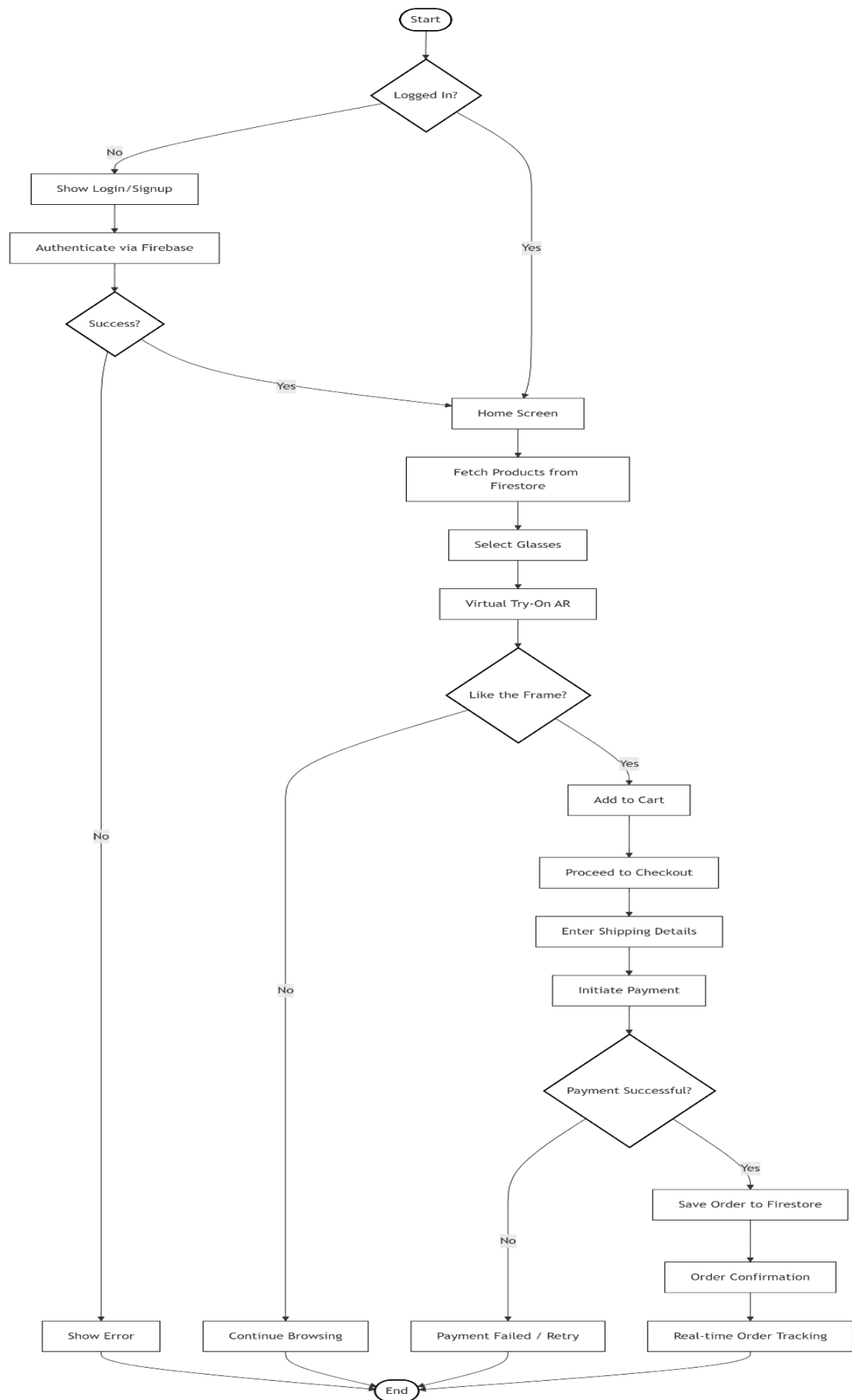


Figure 4:2 Process Flow

The diagram depicts a complete Process Flow for the GAR Optical application. It illustrates the user's journey from opening the app, checking their Login Status (for authentication), navigating through the Catalog and Virtual Try-On (AI/AR) features, and finally completing a transaction through Checkout and Order Confirmation.

4.3 Design Models

The design models for GAR OPTICALS outline the interactions and data structures essential for supporting the platform's core features: product browsing, virtual try-on, personalized recommendation, and secure transactions. These models include sequence and class diagrams to depict the flow of operations and the structure of key components.

4.3.1 Sequence Diagram

The sequence diagram illustrates the process flow for a user engaging with the core innovation of the app: product search, visualization via Augmented Reality (AR), and the decision to purchase.

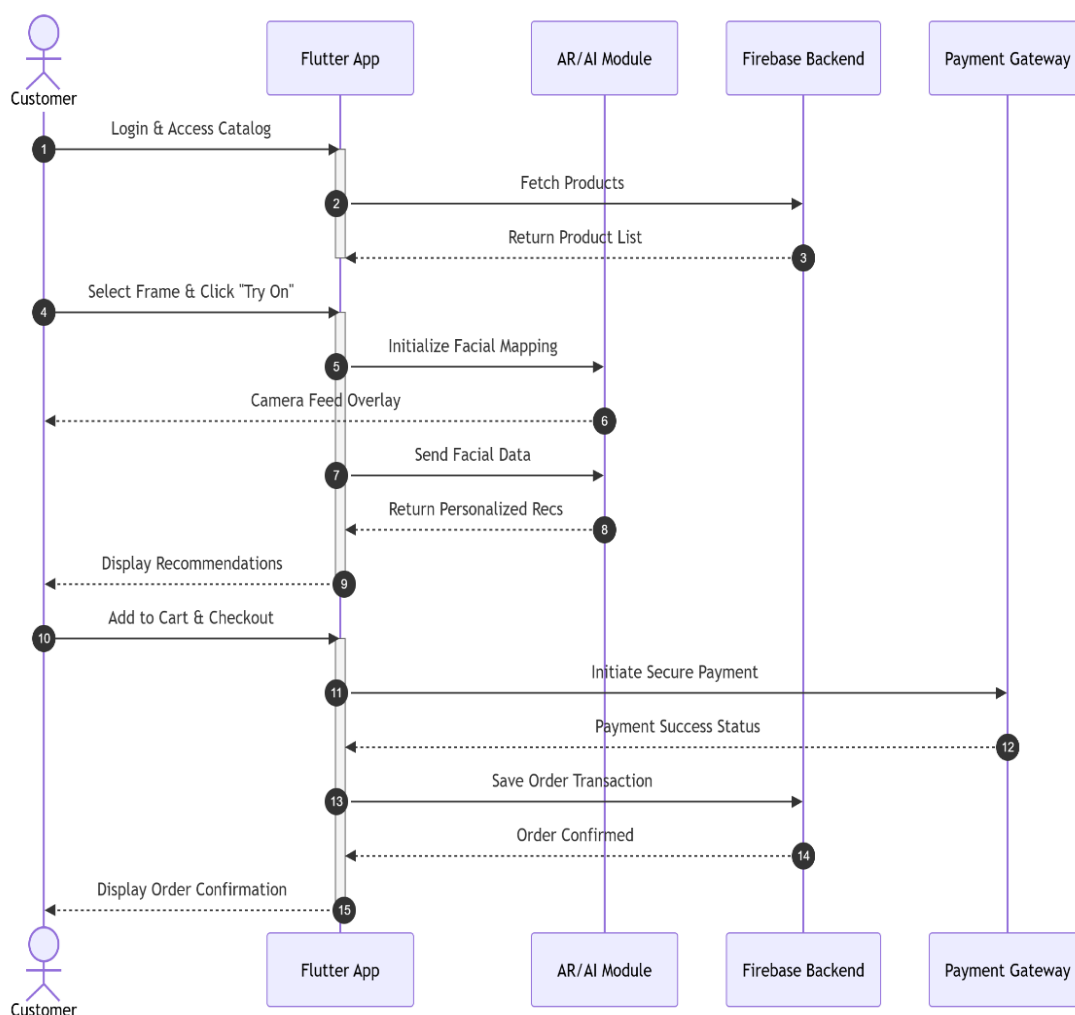


Figure 4:3 Sequence Diagram of GAR

4.3.1.1 Virtual Try-On and Purchase Flow

The sequence begins with the user accessing the platform and initiating the visualization process. The system handles the real-time AR processing and subsequently facilitates the secure transaction if the user decides to buy.

- **User Access:** The user logs in and accesses the home page or catalog.
- **Visualization:** The user selects a frame and initiates the Virtual Try-On feature, allowing the AR Module to capture facial data.
- **Intelligence:** The AR/AI Modules validate the input, overlay the frame, and generate personalized recommendations.
- **Transaction:** If the user adds the product to the cart and initiates payment, the system coordinates with the Payment Gateway to ensure a secure transaction and logs the order details.

4.3.2 Class Diagram

The class diagram represents the core classes (entities) within GAR OPTICALS, detailing their attributes and relationships. This structure defines how user data, product listings, AR assets, and transactions are managed. Key classes include:

- **GarPlatform (Main System Class):** The central controller that initializes the system and coordinates the core features.
- **User:** Stores user-specific information and manages roles (Customer/Admin).
- **ProductListing:** Represents a single frame or lens product available in the catalog.
- **ARAsset:** Manages the 3D models and visualization data required for the AR try-on feature.
- **Recommendation:** Represents the personalized frame suggestions generated by the AI module.
- **Transaction:** Manages the details of completed or pending user purchases.

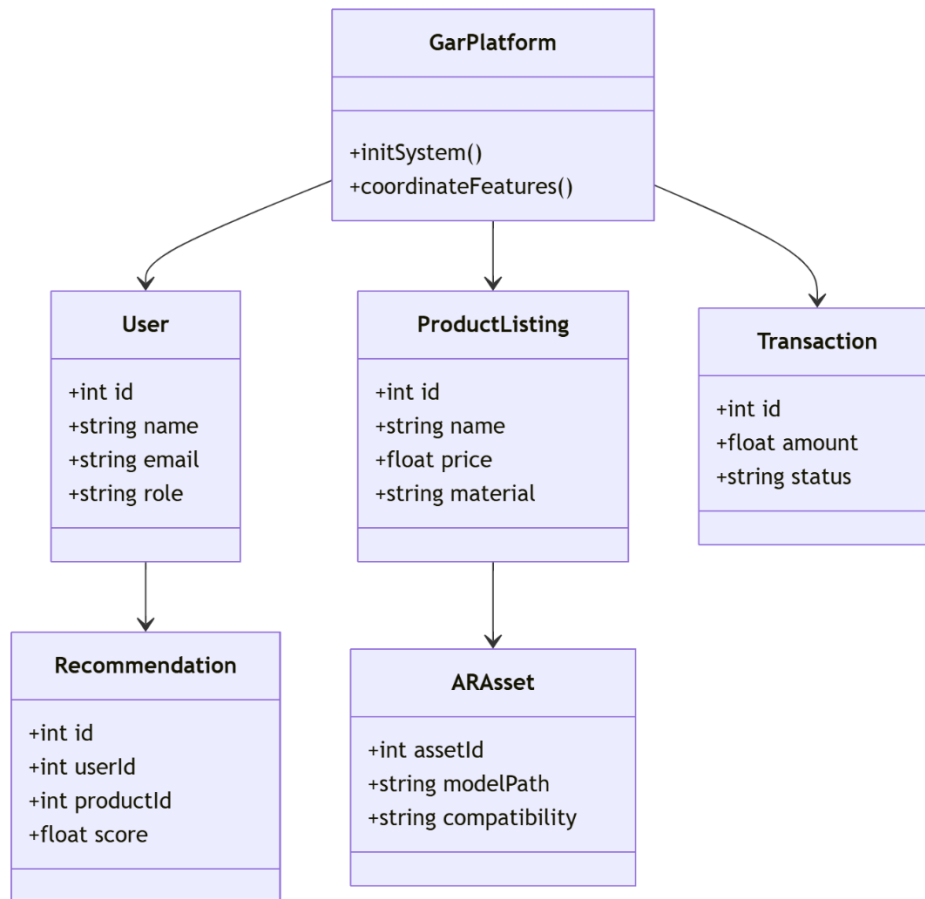


Figure 4:4 Class Diagram

4.3.2.1 User

Stores user information, including ID, name, email, role (customer or admin), and account details.

4.3.2.2 ARAsset

Manages the 3D models and visualization data required for the AR try-on feature, with attributes such as asset ID, model path, and frame compatibility metrics.

4.3.2.3 Recommendation

Represents a suggestion generated by the AI, with attributes such as ID, user ID, suggested product ID, and reason/score.

4.3.2.4 ProductListing

Represents an item in the catalog, with attributes such as ID, name, style, material, price, and a reference to its ARAsset.

4.3.2.5 Transaction

Manages transaction details, including ID, buyer ID, product ID, amount, and payment status

(e.g., pending, complete, failed).

The Class Diagram illustrates the main structural entities of the GAR OPTICALS system, including the central Gar Platform and its relationship with the core modules. Key entities are Product Listing, the ARAsset for visualization, the Recommendation Engine (AI), and the Transaction class, defining the attributes and operations for each component to maintain data integrity and system functionality.

4.4 Data Design

The data design for GAR OPTICALS defines the database structure and the data elements necessary to support the platform's features, including user management, product listings, AR assets, and transaction processing. A data dictionary is used to provide detailed descriptions of each key table and its attributes.

4.4.1 Data Dictionary

The database for GAR OPTICALS is organized into key tables, each representing a fundamental entity within the platform (e.g., Users, Products, Transactions). The primary tables and their attributes are outlined below:

Table 4.1 Data Dictionary

Table	Attributes	Data Type	Key
Users	UserID, Name, Email, Role, Password	Int, String, String, String, String	PK
Product Listings	ListingID, SellerID, Name, Style, Price, ARAssetID, Condition	Int, Int, String, String, Decimal, Int, String	PK, FK
AR Assets	AssetID, FilePath, ModelName, ValidationStatus	Int, String, String, Boolean	PK
Transactions	TransactionID, BuyerID, ProductID, PaymentStatus, Amount	Int, Int, Int, String, Decimal	PK, FK

Table	Attributes	Data Type	Key
Recommendations	RecID, UserID, ProductID, AlgorithmType, Score	Int, Int, Int, String, Float	PK, FK

The inclusion of data types and key constraints in the dictionary provides a standardized technical reference for the implementation phase, ensuring data integrity across the Firebase Firestore database and simplifying the mapping of logical entities to physical storage.

4.4.2 Table Descriptions

4.4.2.1 Users

Stores user credentials and roles, differentiating between customers and administrators.

4.4.2.2 Product Listings

Contains information for each product (glasses/lenses), including style, price, condition, and a reference to the AR model, helping customers assess the product.

4.4.2.3 AR Assets

Stores file paths and validation results for the 3D models used in the Virtual Try-On feature.

4.4.2.4 Transactions

Manages transaction records, including buyer details, product IDs, and payment status, ensuring secure order management.

4.4.2.5 Recommendations

Holds details about the personalized suggestions generated by the AI system, including the algorithm used and the suggestion score.

CHAPTER 5

5. Implementation

This chapter details the technical implementation of the GAR OPTICALS system. It explains the core algorithms used to power the AR and AI features, the integration of external APIs, and the design philosophy behind the user interface.

5.1 Algorithm

In this section, we define the logic for the project's two most critical modules: the AR Try-On engine and the AI-based Recommendation engine.

5.1.1 Virtual Try-On Algorithm (Natural Language)

The AR Try-On functionality relies on real-time face tracking. The logic follows these steps:

- **Initialize Engine:** Request camera permissions; initialize ARCore/ARKit session.
- **Face Detection:** The system continuously scans the camera feed to detect facial landmarks (eye distance, nose bridge, temple width).
- **Frame Alignment:** Based on the detected facial mesh, the system calculates the 3D coordinates (x, y, z) for the frame placement.
- **Overlay:** The selected 3D model is rendered and dynamically adjusted to match the user's head movement and rotation.
- **Session Terminate:** If the user exits the AR screen, the session is cleared to save device memory.

5.1.2 AI Recommendation Logic (Pseudocode)

The recommendation system uses a weighted scoring algorithm to match frames to user profiles:

Plaintext

```
FUNCTION GetPersonalizedRecommendations(UserID, UserFaceShape):
```

```
    Products = Firestore.FetchAll("Products")
```

```
    Recommendations = []
```

```
    FOR Product in Products:
```

```
        IF Product.Style matches UserFaceShape:
```

```
            Score = BaseScore + (Product.Popularity * 0.5)
```

Recommendations.Add(Product, Score)

Sort Recommendations by Score DESC

RETURN Recommendations.Top(5)

5.2 External APIs

The system integrates several third-party APIs to handle specialized tasks like payments, data storage, and intelligent conversational support.

Table 5.1 Details of APIs used in the project

Name of API	Description of API	Purpose of Usage	List down the function/class name in which it is used
Firebase Auth	Identity management service	Secure User Login/Signup	FirebaseAuth.instance.signInWithEmail()
Cloud Firestore	NoSQL Cloud Database	Storing Products, Orders, & User Data	FirebaseFirestore.instance.collection()
Dialogflow	Google's Conversational AI	Customer Support Chatbot	DialogflowDetector.detectIntent()
Stripe/Razorpay	Payment Processing SDK	Handling secure transactions	PaymentGateway.processTransaction()
ARCore/ARKit	Augmented Reality SDK	Facial mapping & Frame overlay	ArCoreController.addNode()

5.3 User Interface

The GAR OPTICALS interface is designed to follow modern mobile design standards, focusing on clarity, speed, and user engagement.

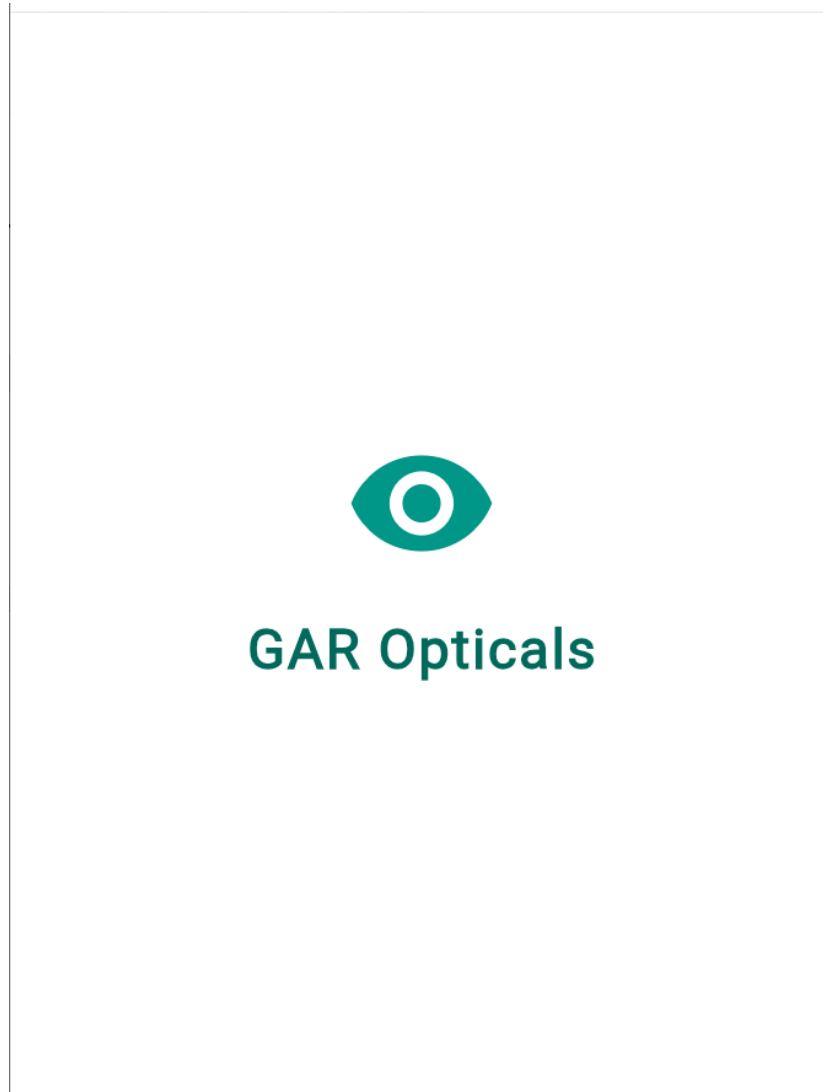


Figure 5.1 UI of Splash Screen

This figure illustrates the initial splash and branding screen of the GAR Optical mobile application, which serves as the primary entry point for users. The interface features a clean, minimalist design centered around the brand's logo and title to establish a professional visual identity. This screen is designed to load essential application assets and initialize the Firebase Authentication state in the background. By providing a smooth transition into the main functional modules, it enhances the overall user experience and brand recognition. The color scheme and typography are chosen to align with the modern, tech-driven aesthetic of the Virtual Try-On system.

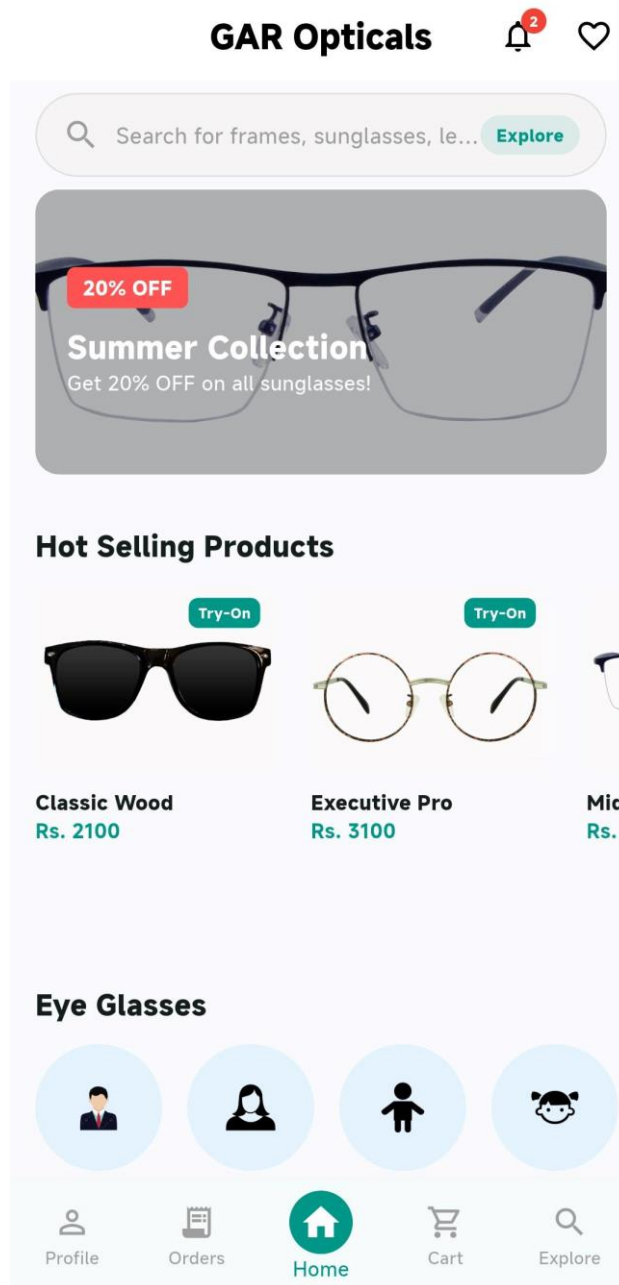


Figure 5.2 UI of Home Screen – Hot Selling Products

This figure displays the primary user interface of the GAR Optical's home screen, designed to provide a comprehensive and engaging shopping experience. The screen features a prominent search bar and a dynamic promotional banner to highlight active discounts and new collections to the user. Below the banner, a "Hot Selling Products" section showcases top eyewear items with high-quality images and clear pricing, fetched directly from the Firestore database. The bottom navigation bar allows for seamless transitions between the user's profile, order history, and the virtual try-on exploration modules. This layout is optimized for high usability, ensuring that customers can easily navigate the product catalog and identify trending styles.

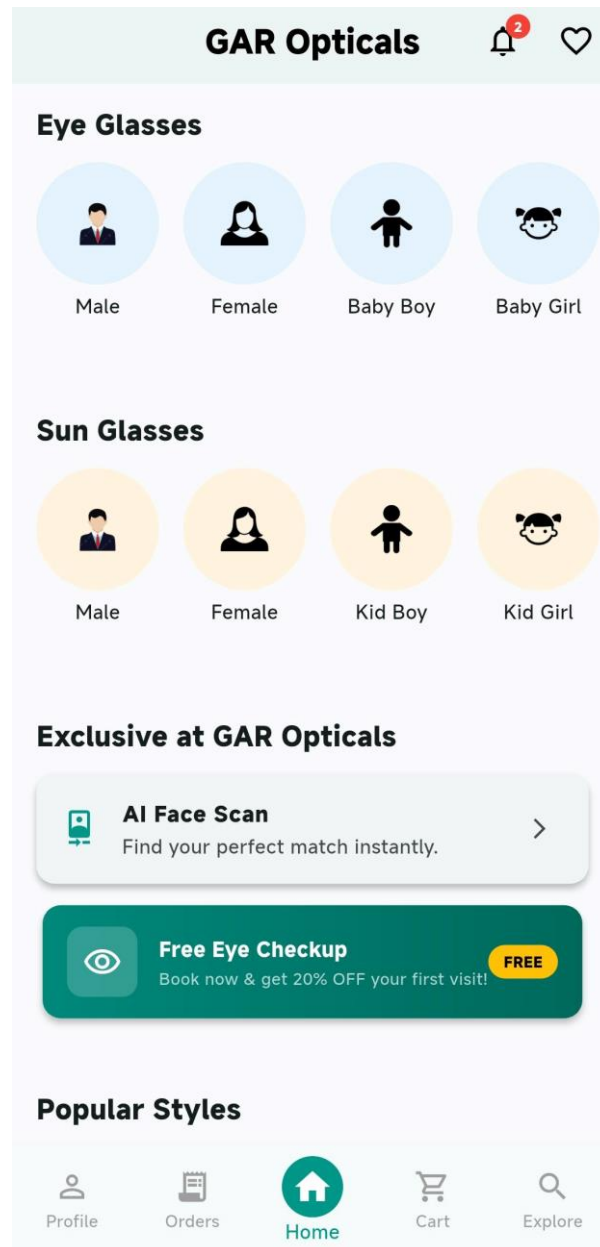


Figure 5.3 UI of Home Screen – Categories and Exclusive Offers

This figure presents the secondary section of the GAR Optical's home screen, which focuses on targeted category selection and exclusive user features. The interface organizes products by demographic segments under dedicated sections for "Eye Glasses" and "Sun Glasses," allowing users to instantly filter items for men, women, and children. Below these categories, the "Exclusive at GAR Optical's" panel introduces specialized services, including an interactive "AI Face Scan" feature to help customers find their ideal frame match automatically. Additionally, a promotional banner highlights a "Free Eye Checkup" booking option to drive customer engagement and service utilization.

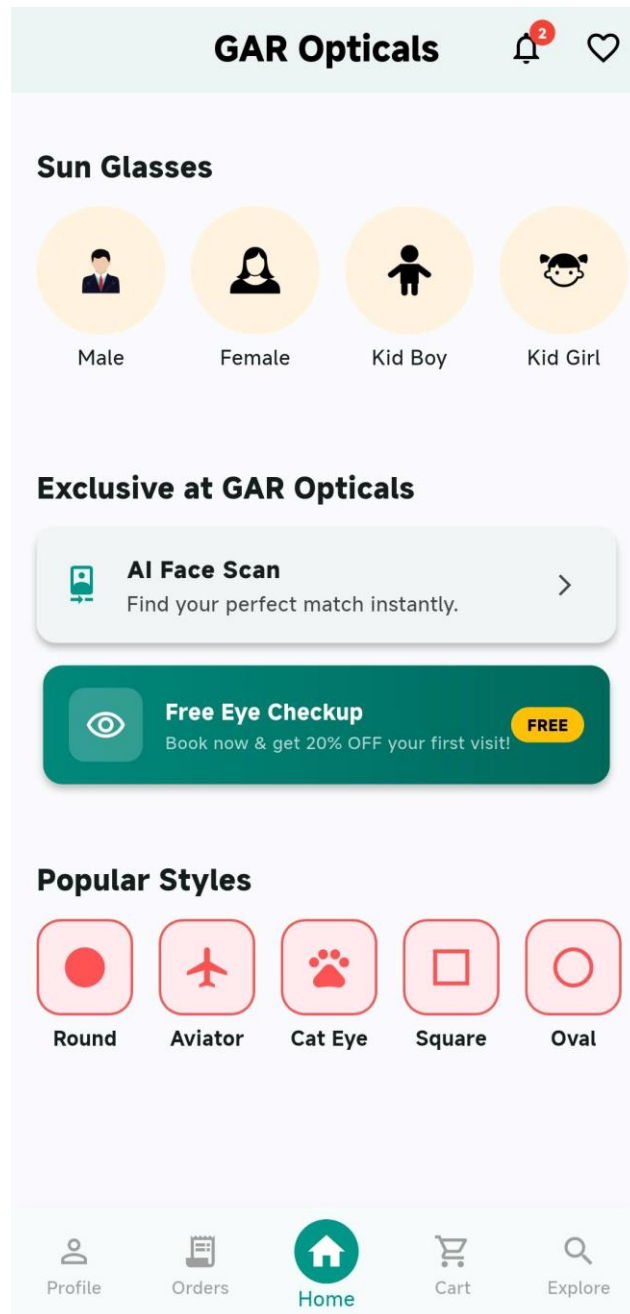


Figure 5.4 UI of Home Screen – Popular Styles

This figure illustrates the lower segment of the GAR Optical's home screen, focusing on style-based filtering and specialized user engagement modules. The interface features a "Popular Styles" section that enables users to browse eyewear based on specific geometric frame shapes, including Round, Aviator, Cat Eye, Square, and Oval categories. Positioned directly above this is the "Exclusive at GAR Optical's" panel, which highlights premium features such as an automated "AI Face Scan" for instant frame matching and a promotional "Free Eye Checkup" booking card.

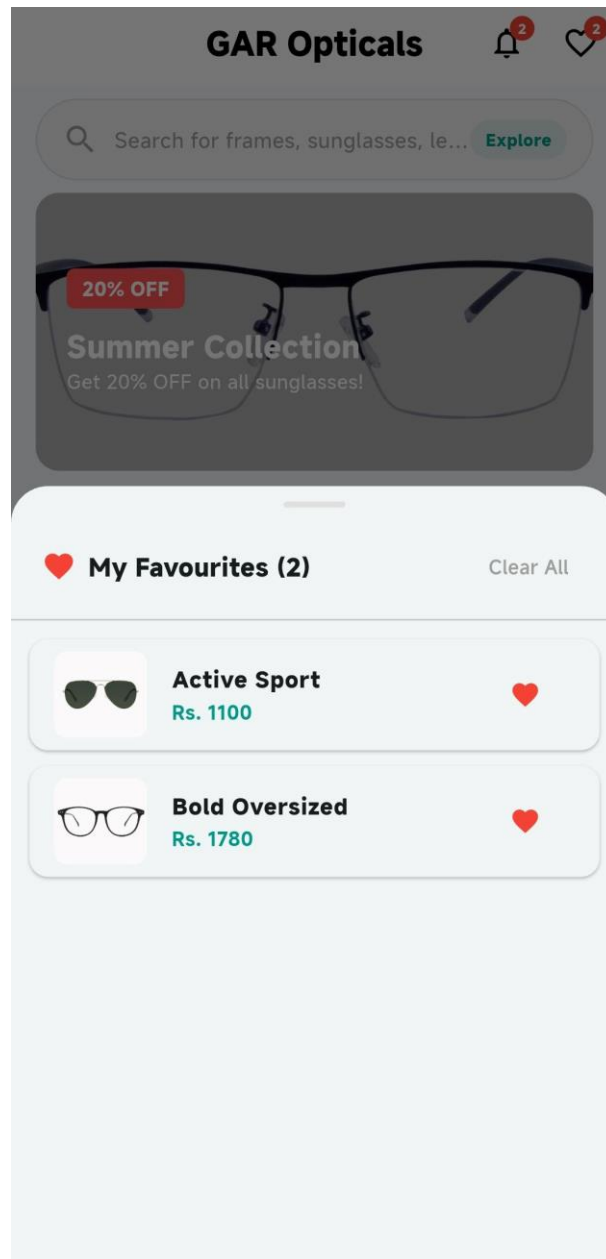


Figure 5.5 UI of My Favourites Screen

This figure displays the "My Favourites" bottom sheet modal within the GAR Optical's application, allowing users to quickly view and manage their shortlisted eyewear items. The modal overlays the main home screen and dynamically lists favorited products, such as "Active Sport" and "Bold Oversized," along with their respective prices fetched from the backend. Each item card features a product thumbnail and an interactive heart icon, enabling users to effortlessly remove items from their wishlist. Additionally, a "Clear All" option is provided at the top right of the sheet for bulk list management. This modular overlay enhances the user experience by providing instant access to saved preferences without forcing the user to navigate away from their current page.

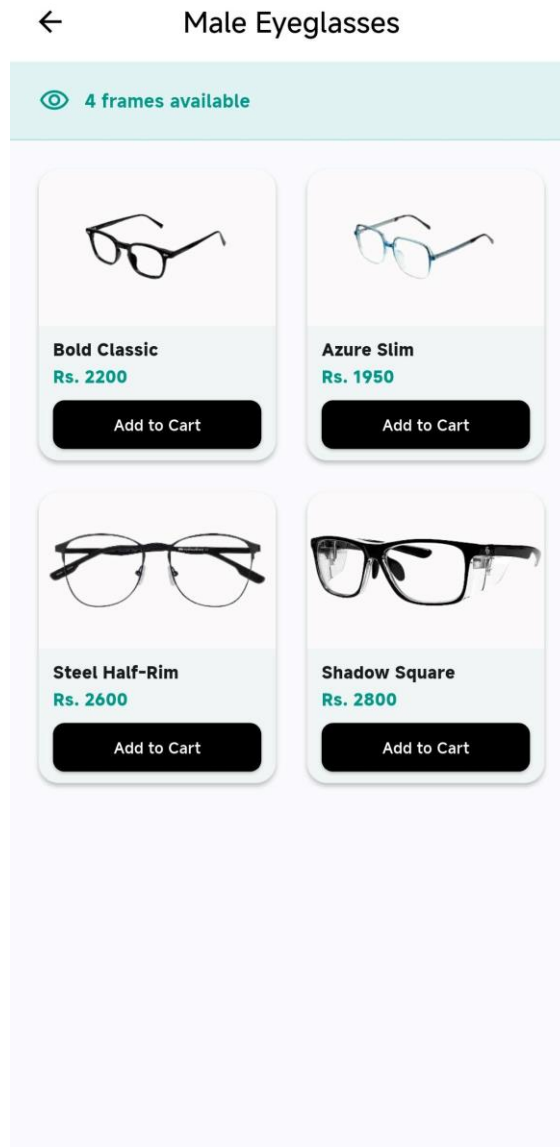


Figure 5.6 UI of Male Eyeglasses Listing Screen

This figure showcases the dedicated product listing page for the "Male Eyeglasses" category within the GAR Opticals application. The interface utilizes a clean, two-column grid layout to present available products, displaying key details such as high-quality product images, model names (e.g., "Bold Classic", "Azure Slim"), and accurate pricing information retrieved dynamically from the database. A status banner at the top of the grid indicates the total number of frames available in the current view, keeping users informed as they browse. Each product card includes an prominent, accessible "Add to Cart" button to streamline the purchasing workflow. The top app bar features an intuitive back navigation arrow alongside clear screen labeling, ensuring a straightforward and user-friendly catalog browsing experience.

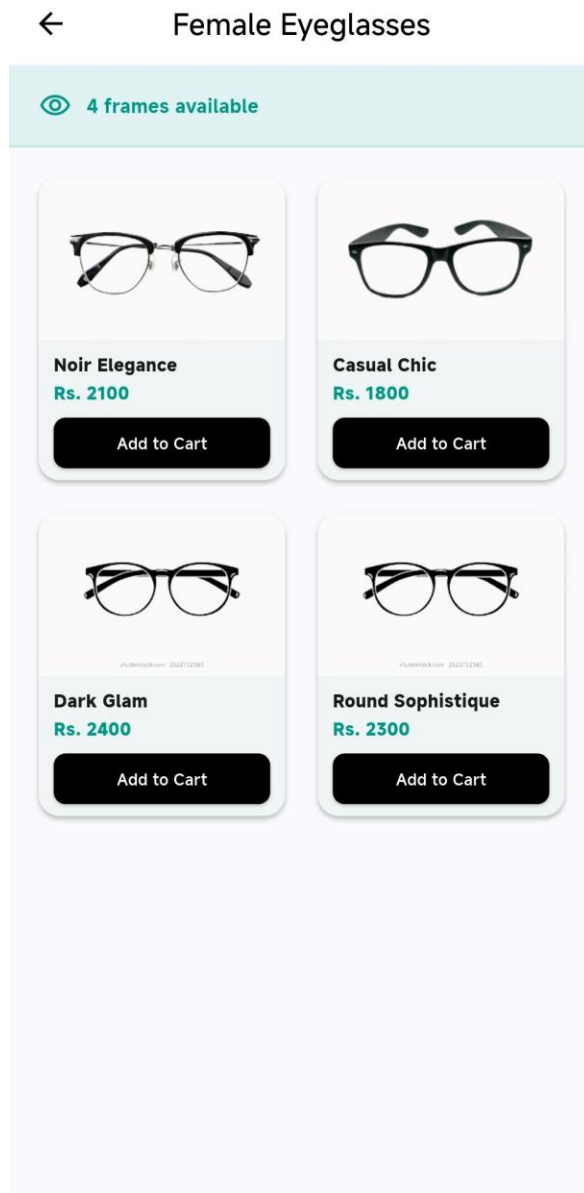


Figure 5.7 UI of Female Eyeglasses Listing Screen

This figure presents the dedicated product catalog interface for the "Female Eyeglasses" category in the GAR Opticals application. Similar to the male category layout, it implements a highly responsive, two-column grid system to showcase individual eyewear options such as "Noir Elegance" and "Casual Chic." Each product card dynamically renders high-definition product graphics alongside pricing metrics and model titles retrieved from the backend system. An indicator bar at the top displays the total available inventory count in real time, helping to streamline user expectations during browsing. A bold "Add to Cart" call-to-action is integrated seamlessly onto every card module to expedite user checkout workflows, while an intuitive back navigation button in the header ensures trouble-free navigation.

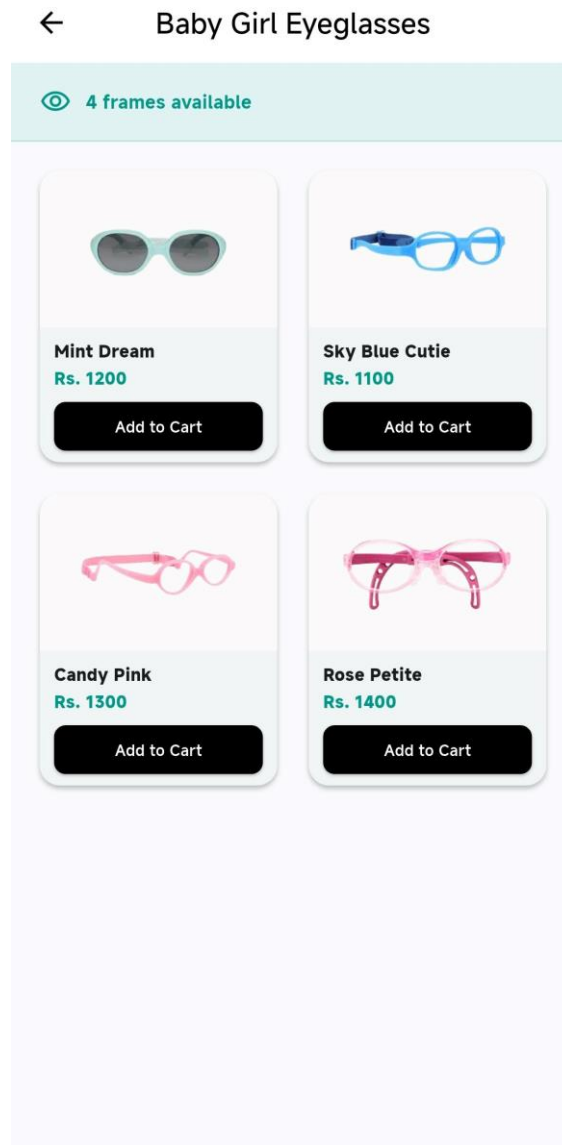


Figure 5.8 UI of Baby Girl Eyeglasses Listing Screen

This figure displays the specialized category listing interface for "Baby Girl Eyeglasses" within the GAR Opticals application. The screen maintains the application's clean design signature, arranging child-friendly eyewear models such as "Mint Dream" and "Sky Blue Cutie" into a balanced, two-column grid layout. An indicator bar sits beneath the header to show that 4 frames are currently available, matching the inventory count structure used throughout the product modules. Each unique product card displays a vibrant thumbnail image, dynamic pricing data fetched directly from the database, and a distinct "Add to Cart" button to streamline the checkout process. A simple back navigation button is integrated into the top app bar, allowing users to return to the primary dashboard seamlessly.

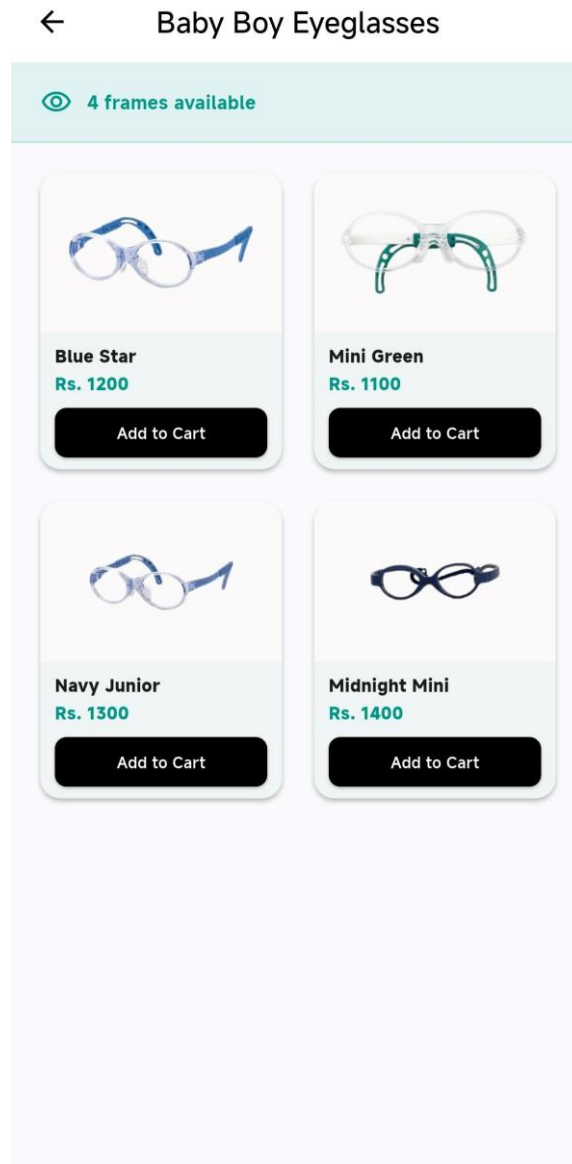


Figure 5.9 UI of Baby Boy Eyeglasses Listing Screen

This figure presents the dedicated product catalog interface for the "Baby Boy Eyeglasses" category within the GAR Opticals application. The screen employs a clear, structured two-column grid layout to display a collection of child-centric eyewear variants such as "Blue Star" and "Mini Green." An upper informational strip dynamically tracks inventory availability, indicating that 4 frames are currently available for selection. Each standalone product block integrates high-resolution asset imagery, localized pricing pulled directly from the system backend, and an action-oriented "Add to Cart" element to optimize the customer conversion funnel. A functional back arrow resides within the header area to ensure intuitive navigation back to the primary store categories.

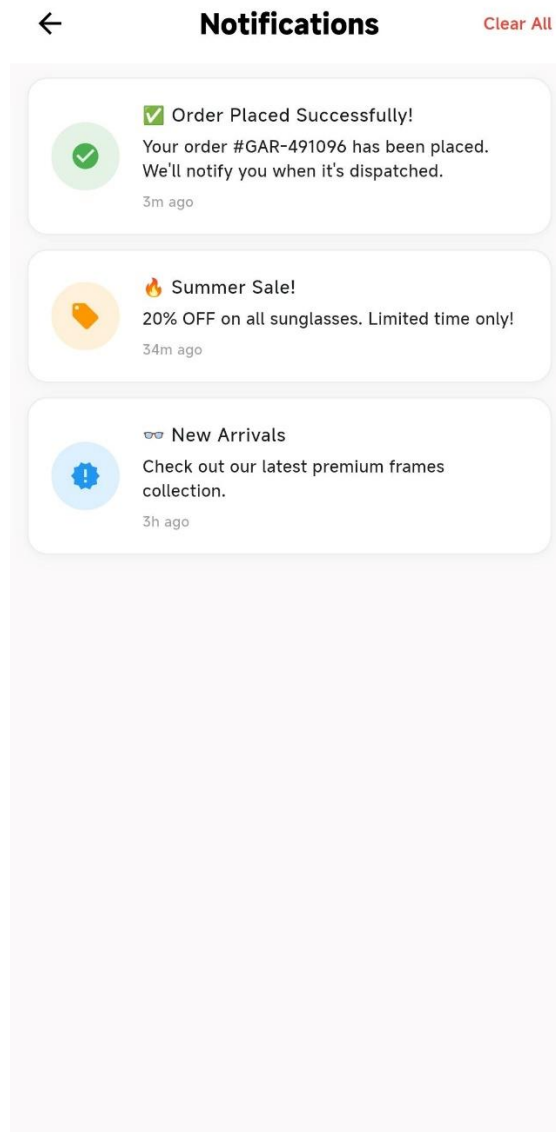
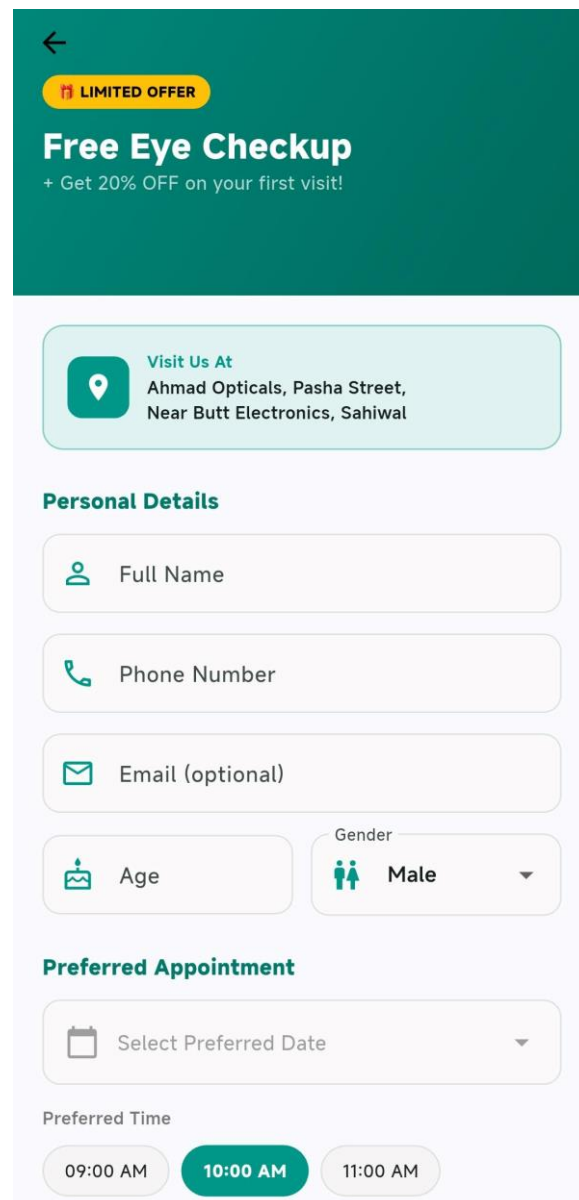


Figure 5.10 UI of Notifications Screen

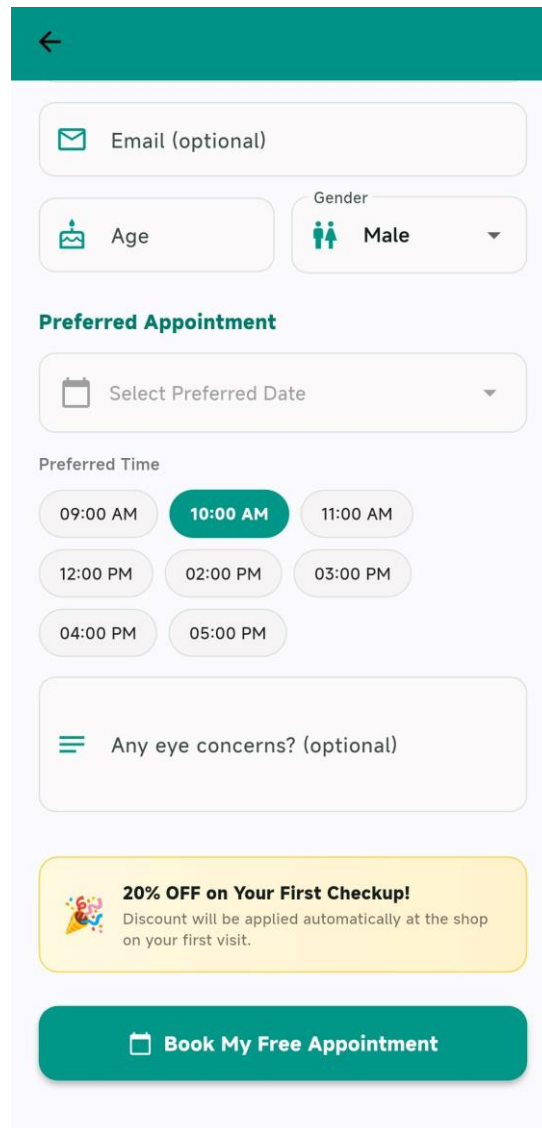
This figure illustrates the user "Notifications" center screen within the GAR Opticals application, designed to keep users informed about order updates and promotional marketing events. The list view features structured notification blocks, such as transactional confirmations (e.g., "Order Placed Successfully!") alongside global marketing announcements for seasonal promotions and new product arrivals. Each notification module includes context-specific icons, explicit text summaries, and dynamic time-ago indicators to reflect the real-time nature of the updates. A prominent "Clear All" utility button is positioned in the top right header to give users full organizational control over their alerts list. The top bar also maintains a standard back-navigation arrow button, ensuring smooth transition back to the main app dashboard.



The image shows a mobile application interface for booking a free eye checkup. At the top, there is a dark green header with a back arrow, a yellow 'LIMITED OFFER' badge, and the text 'Free Eye Checkup' followed by '+ Get 20% OFF on your first visit!'. Below the header is a light blue location card with a map icon and the text 'Visit Us At Ahmad Opticals, Pasha Street, Near Butt Electronics, Sahiwal'. The main form area is white and contains two sections: 'Personal Details' and 'Preferred Appointment'. The 'Personal Details' section has input fields for 'Full Name', 'Phone Number', and 'Email (optional)'. It also has a row with 'Age' (with a birthday icon) and 'Gender' (a dropdown menu currently showing 'Male'). The 'Preferred Appointment' section has a date selector labeled 'Select Preferred Date' and a 'Preferred Time' section with three buttons: '09:00 AM', '10:00 AM' (which is highlighted in green), and '11:00 AM'.

Figure 5.11 UI of Free Eye Checkup Booking Screen

This figure demonstrates the "Free Eye Checkup" booking form interface within the application, designed to streamline clinical appointment scheduling for users. The top header explicitly features a promotional banner outlining the offer and an associated discount incentive. Below the header, a dedicated location card details the physical branch address to ensure clear navigational guidance for physical visits. The main form area contains a "Personal Details" section equipped with clean interactive input fields for capturing user information such as full name, contact number, email, age, and gender. Furthermore, a "Preferred Appointment" module is integrated into the lower layout, allowing users to choose an available date from a dropdown selector and choose from discrete preferred time slots to complete their booking.



The image shows a mobile application interface for booking a free eye checkup. At the top, there is a teal header with a white back arrow. Below the header, there is a form with several input fields: "Email (optional)" with an envelope icon, "Age" with a birthday icon, and "Gender" with a dropdown menu showing "Male" and a person icon. Below these fields, there is a section titled "Preferred Appointment" in teal. This section contains a date picker labeled "Select Preferred Date" and a "Preferred Time" section with a grid of time slots. The time slots are: 09:00 AM, 10:00 AM (selected and highlighted in teal), 11:00 AM, 12:00 PM, 02:00 PM, 03:00 PM, 04:00 PM, and 05:00 PM. Below the time slots, there is a text input field labeled "Any eye concerns? (optional)" with a list icon. At the bottom of the form, there is a yellow promotional banner that says "20% OFF on Your First Checkup!" with a gift icon and a note that the discount will be applied automatically at the shop on the first visit. Finally, at the very bottom, there is a large teal button with a white calendar icon and the text "Book My Free Appointment".

Figure 5.12 UI of Free Eye Checkup Booking Screen – Submission

This figure displays the lower segment of the appointment scheduling screen within the GAR Optical application, highlighting the finalized booking details and submission interface. The view extends the data entry fields by providing an optional multi-line text container labeled "Any eye concerns?" for patients to describe specific symptoms prior to their visit. Beneath the input controls, a promotional banner explicitly informs the user that a 20% discount incentive will be applied automatically at the storefront upon their initial visit. The interface concludes with a prominent, full-width "Book My Free Appointment" action button that triggers validation workflows and sends appointment data directly to the database backend. A standardized header configuration featuring an intuitive back arrow remains accessible at the top left to ensure uninhibited user navigation.

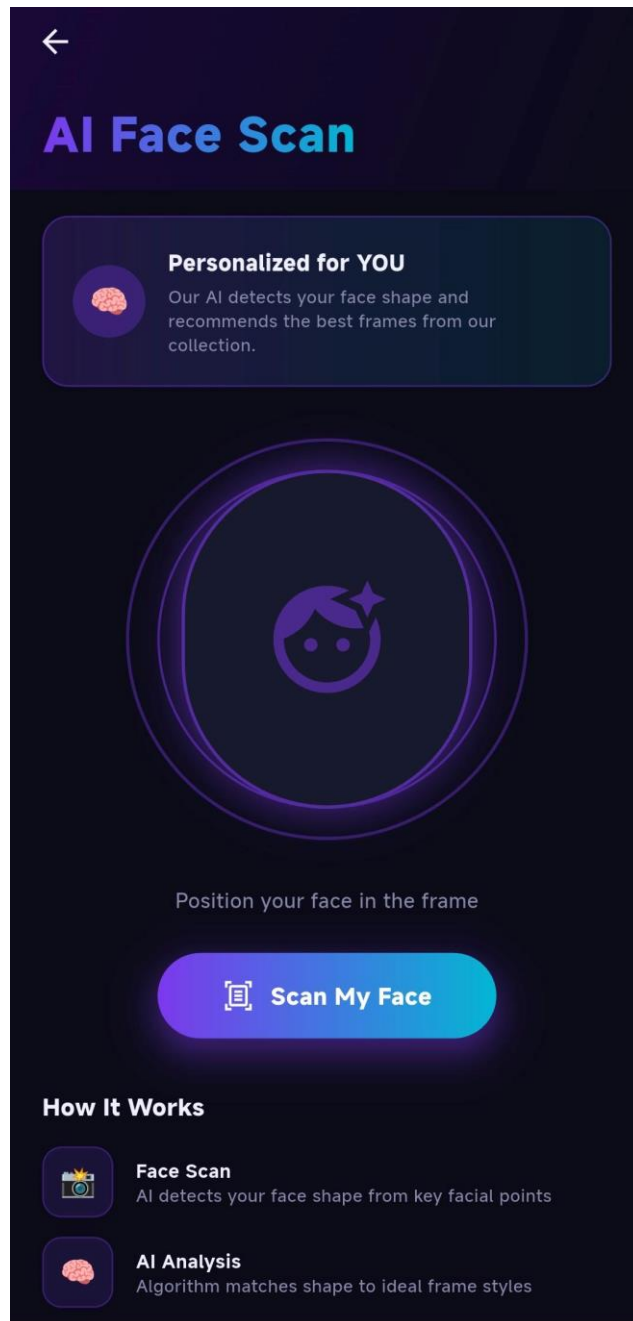


Figure 5.13 UI of AI Face Scan Screen

This figure presents the interactive "AI Face Scan" feature screen within the application, utilizing a dark-themed futuristic aesthetic to distinguish its advanced technological functionalities. The top header block houses an introductory banner explaining that the system detects face shapes to provide personalized frame recommendations. Centered on the screen is a prominent, circular scanning viewfinder widget with an explicit overlay directive instructing users to position their face within the frame. Below this scanner module, a vibrant, gradient-styled "Scan My Face" action button initiates the facial mapping sequence. The lower layout concludes with a "How It Works" informational guide that breaks down the underlying logic into two sequential stages.

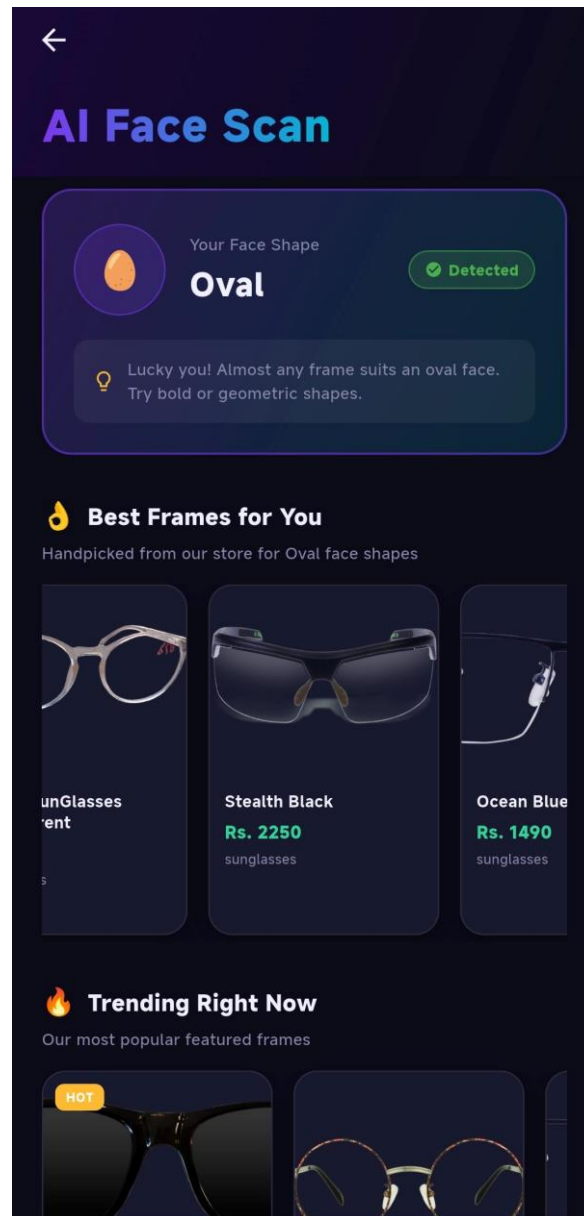


Figure 5.14 UI of AI Face Scan Results Screen

This figure displays the analytical results interface generated following an interactive scan within the GAR Opticals application. The top panel highlights the user's computed biometric profile, explicitly confirming a "Detected" status alongside an "Oval" face shape classification and tailored styling recommendations. Directly below this result module, a dynamic horizontal carousel labeled "Best Frames for You" presents automated product recommendations fetched from the database backend, displaying matching items like "Stealth Black" with localized pricing structures. The lower section introduces a separate "Trending Right Now" product feed featuring popular high-demand frame models. This comprehensive layout maximizes conversion by linking advanced algorithmic computer vision directly to immediate, personalized shopping opportunities.

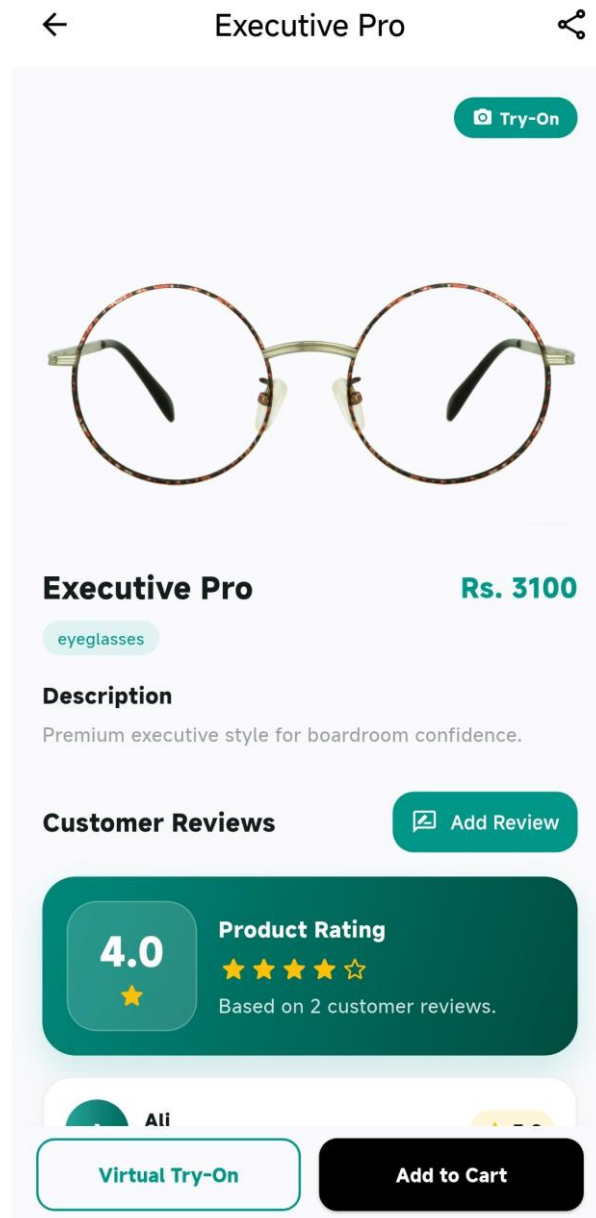


Figure 5.15 UI of Product Detail Screen

This figure presents the detailed product information view for a specific eyewear item, "Executive Pro," within the GAR Opticals application. The upper layout features a high-resolution interactive product image overlaid with a dedicated "Try-On" quick-access badge. Below the asset media, essential product information including the dynamic price point, item category, and a concise textual description are clearly organized. A specialized "Customer Reviews" segment integrates a detailed "Product Rating" block card showing aggregate scores fetched from the database backend alongside a functional "Add Review" button module. The interface concludes with a persistent bottom utility bar offering distinct secondary and primary calls-to-action for "Virtual Try-On" and "Add to Cart" respectively to enhance consumer purchasing workflows.

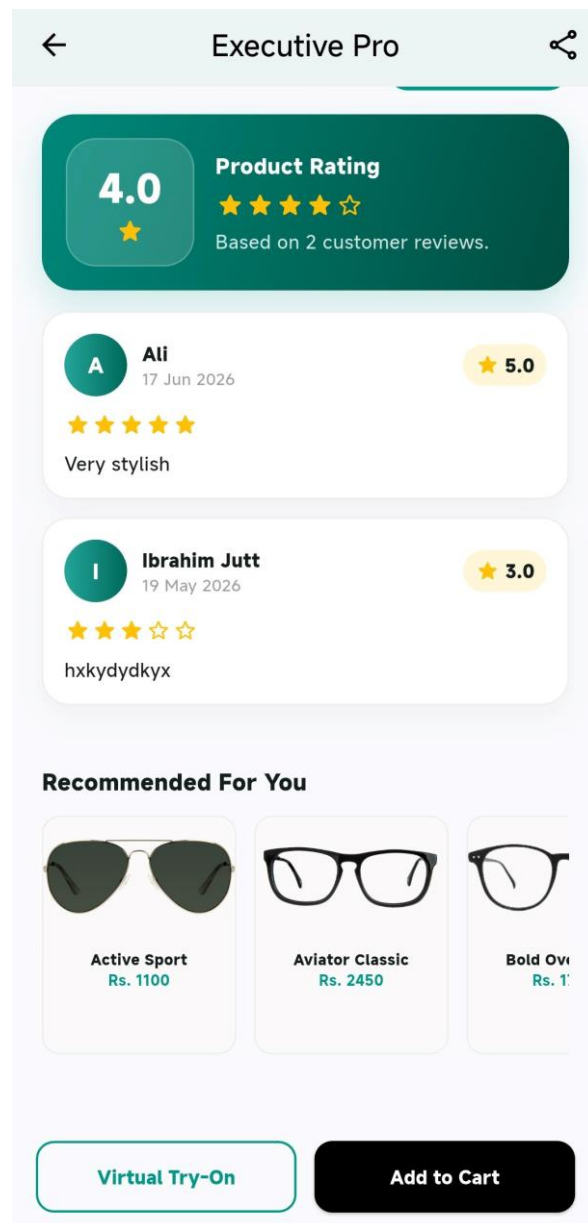


Figure 5.16 UI of Product Detail Screen – Customer Reviews

This figure illustrates the lower section of the "Executive Pro" product screen, highlighting user generated feedback and personal catalog suggestions. The top segment houses individual, scrollable customer review cards containing personalized profile icons, reviewer names, structured star ratings, transaction dates, and specific user commentary fetched directly from the database. Beneath this social proof module, a "Recommended For You" horizontal product carousel displays personalized eyewear alternatives, such as "Active Sport" and "Aviator Classic," alongside their respective prices to promote cross-selling. The screen concludes with the persistent action footer layout, keeping the "Virtual Try-On" and "Add to Cart" interaction buttons fully accessible to the customer during their review evaluation.

My Orders

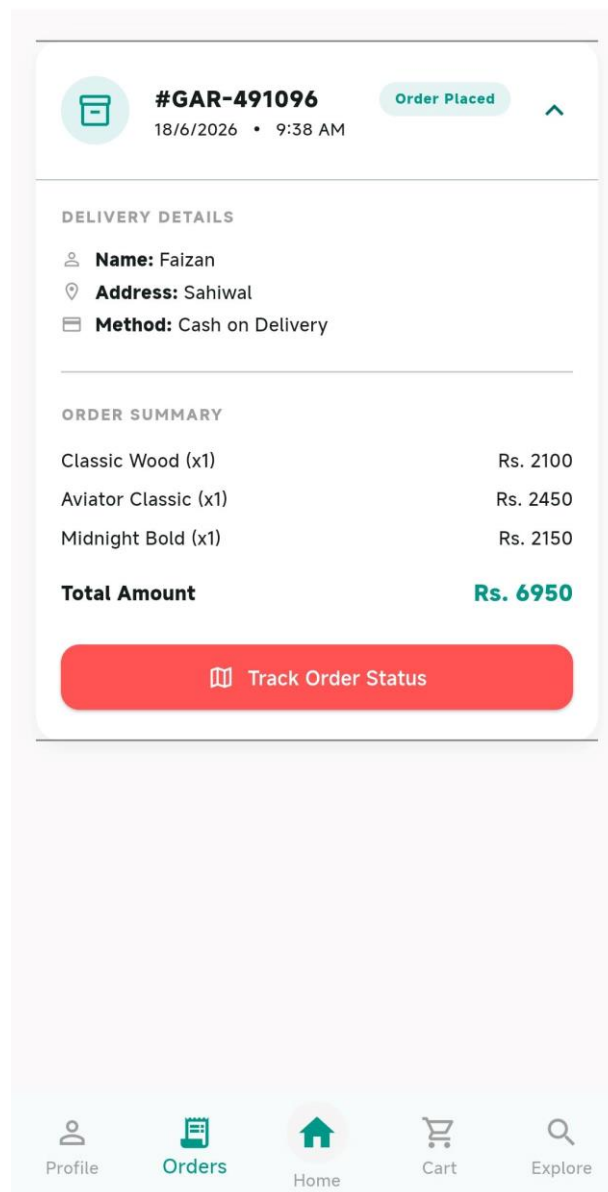


Figure 5.17 UI of My Orders Screen

This figure illustrates the "My Orders" transaction history screen within the GAR Opticals application, accessible via the bottom navigation bar. The interface displays an expandable order card fetched directly from the backend database, showcasing structural order metadata such as the unique Order ID, transaction timestamp, and current fulfillment status badge. Within the expanded card layout, specific modules group key information, including customer delivery metrics (recipient name, shipping address, and payment method) and a detailed itemized order summary with individual unit pricing. A prominent "Track Order Status" action button routes users to the corresponding live tracking timeline stepper to monitor active logistics.

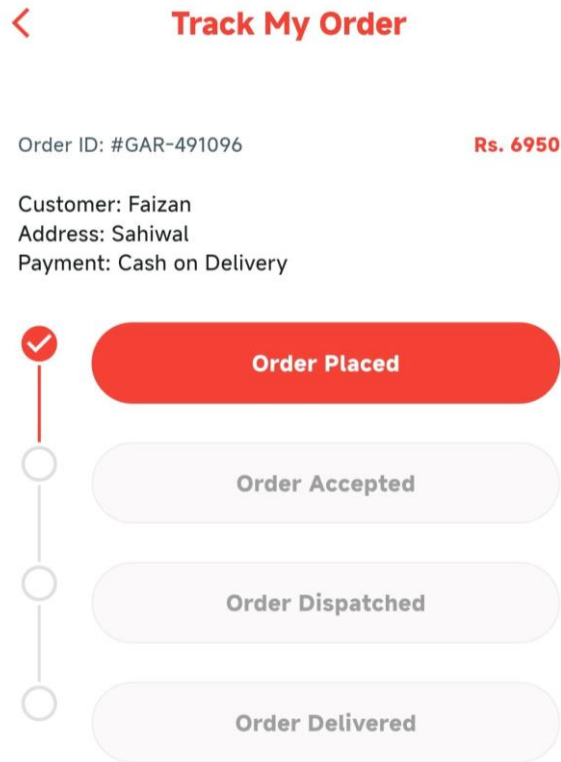


Figure 5.18 UI of Track My Order Screen

This figure displays the "Track My Order" screen within the GAR Opticals application, designed to give users real-time visibility into their purchase fulfillment status. The top layout lists specific invoice data fetched from the backend database, including a unique Order ID, total amount, customer name, shipping address, and the selected payment method. Below these transaction details, a vertical timeline stepper widget dynamically visualizes the processing workflow through four distinct milestones: Order Placed, Order Accepted, Order Dispatched, and Order Delivered. A status indicator and highlighted button container track the current stage of delivery, updating in real time as administrators adjust order records within the administrative backend database.

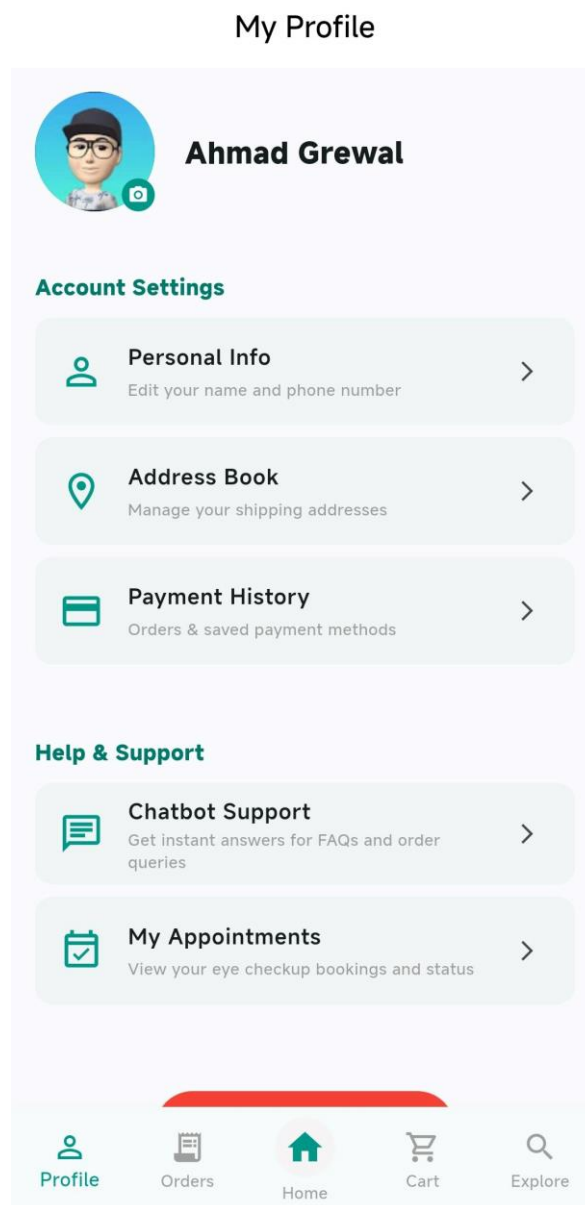


Figure 5.19 UI of My Profile Screen

This figure presents the "My Profile" dashboard layout within the GAR Opticals application, which functions as the central hub for user account management and customer service interactions. The upper portion displays a personalized user profile avatar alongside the authenticated account username. Below the header segment, individual configuration sections group system settings into intuitive modules: "Account Settings" features options to modify personal credentials, update shipping logs via an "Address Book," and monitor "Payment History." Additionally, a dedicated "Help & Support" module contains direct navigational routes to an automated "Chatbot Support" tool for real-time customer queries and a "My Appointments" panel to track clinical booking statuses. The persistent bottom navigation panel highlights the active profile tab, allowing users to move fluidly between core application functionalities.

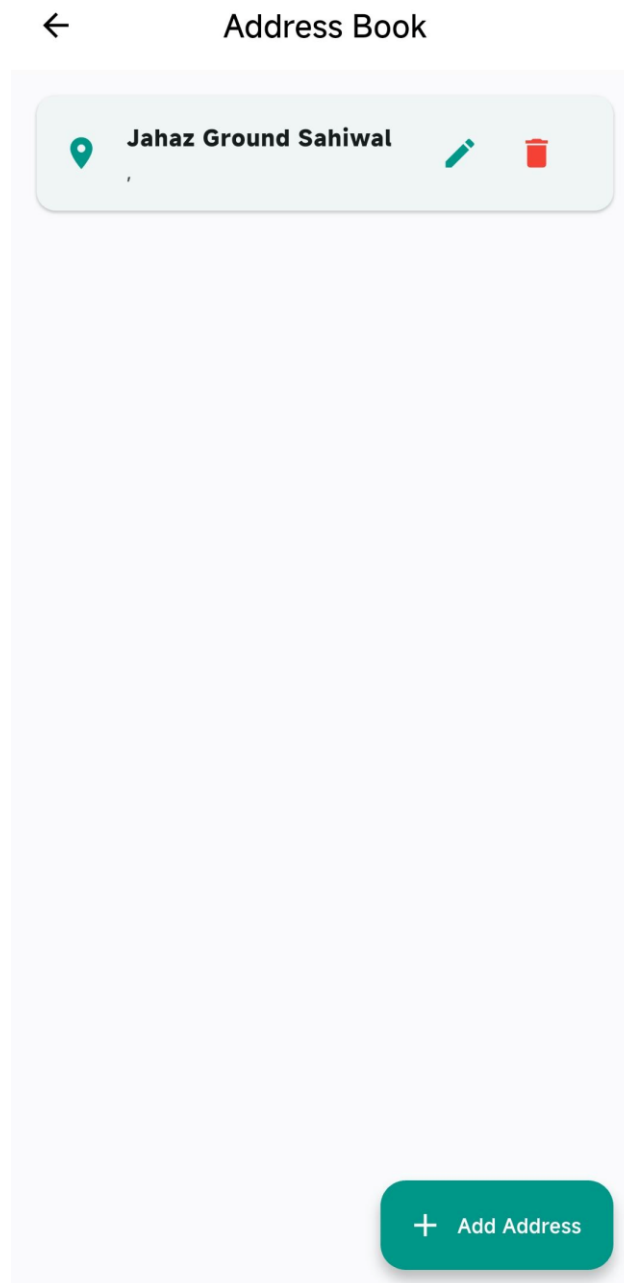


Figure 5.20 UI of Address Book Screen

This figure displays the "Address Book" configuration screen within the GAR Opticals application, which manages a user's saved shipping destinations for checkout acceleration. The main content area features a dynamically populated list of saved address cards fetched from the database, such as "Jahaz Ground Sahiwal," accompanied by a location pin indicator icon. Each entry card integrates contextual action buttons on the right margin, allowing users to instantly edit location details via a pencil icon or remove obsolete records using a red trash bin icon. Additionally, a floating action button labeled "+ Add Address" rests in the lower right corner, providing a clear navigational path to a modal form where users can input and append new delivery metrics to their profile.

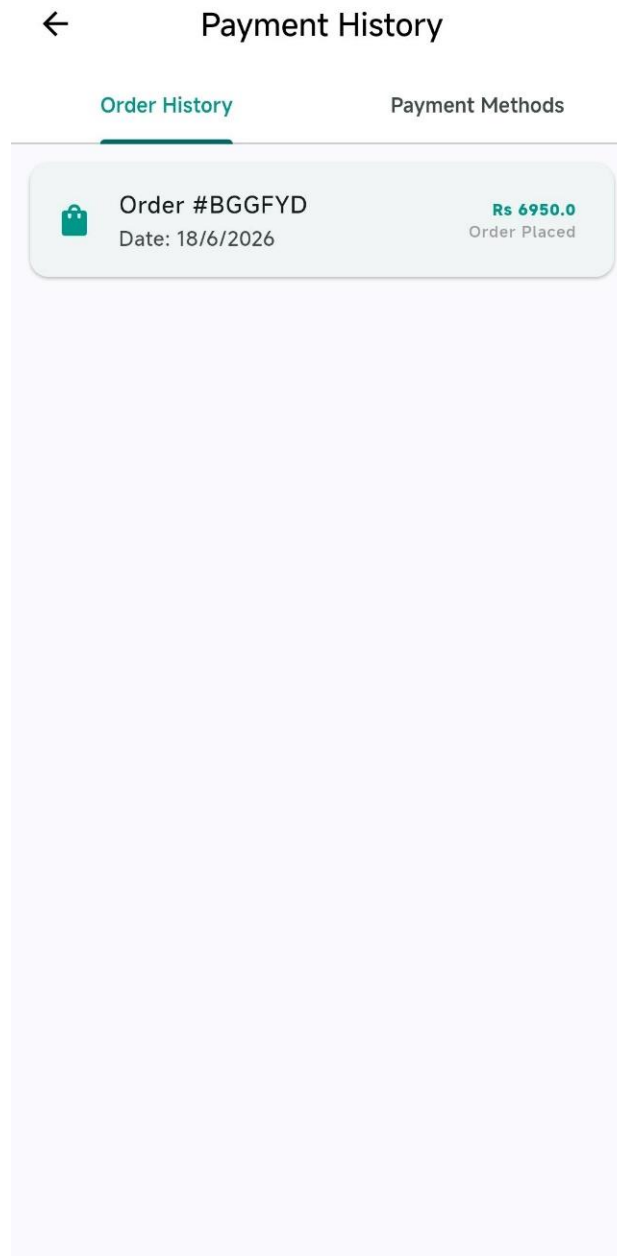


Figure 5.21 UI of Payment History Screen

This figure presents the "Payment History" interface within the GAR Opticals application, allowing users to track financial transactions and billing records. The design features a top tab navigation controller that splits the interface into two main views: "Order History" and "Payment Methods." Under the active "Order History" tab, the interface dynamically populates historical payment summary cards fetched directly from the backend database. Each card itemizes crucial financial and logistical records for a specific transaction, including a unique order reference code, a localized date timestamp, the final transaction value, and a small subtext indicator highlighting the corresponding order fulfillment status.

← Personal Info

Full Name
Ahmad Grewal

Phone Number

Email
ahmadgrewal548@gmail.com

SAVE CHANGES

Figure 5.22 UI of Personal Info Screen

This figure demonstrates the "Personal Info" configuration interface within the GAR Opticals application, which allows users to view and update their primary profile parameters. The layout incorporates structured, outlined text input form fields equipped with corresponding prefix status icons for modifying core identity logs, including full name, contact number, and email address records. These input components are programmatically bound to the backend database, dynamically pre-populating with existing user metrics upon initialization. A prominent "SAVE CHANGES" action button at the base of the form triggers input validation routines and executes asynchronous write operations to commit the updated data profile back to the cloud database storage.

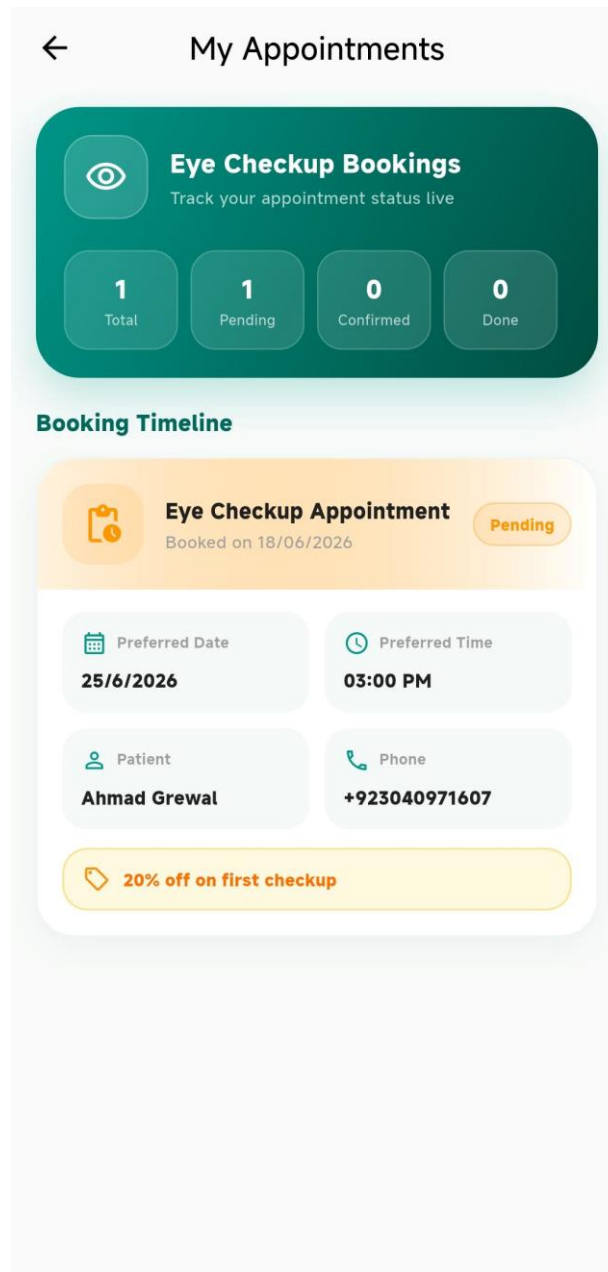


Figure 5.23 UI of My Appointments Screen

This figure showcases the "My Appointments" tracking dashboard within the GAR Opticals application, designed to give users real-time visibility into their eye checkup bookings. The top header block features a statistical analytical banner that aggregates booking statuses from the cloud database, displaying total, pending, confirmed, and completed counts in structured metrics cards. Below this summary, the "Booking Timeline" dynamically renders explicit appointment records, such as an upcoming card showing a "Pending" verification status chip. Individual parameter modules within the card itemize crucial reservation logistics fetched from the backend, including the selected preferred date, preferred time, patient name, contact number, and applicable promotional discounts applied during booking.

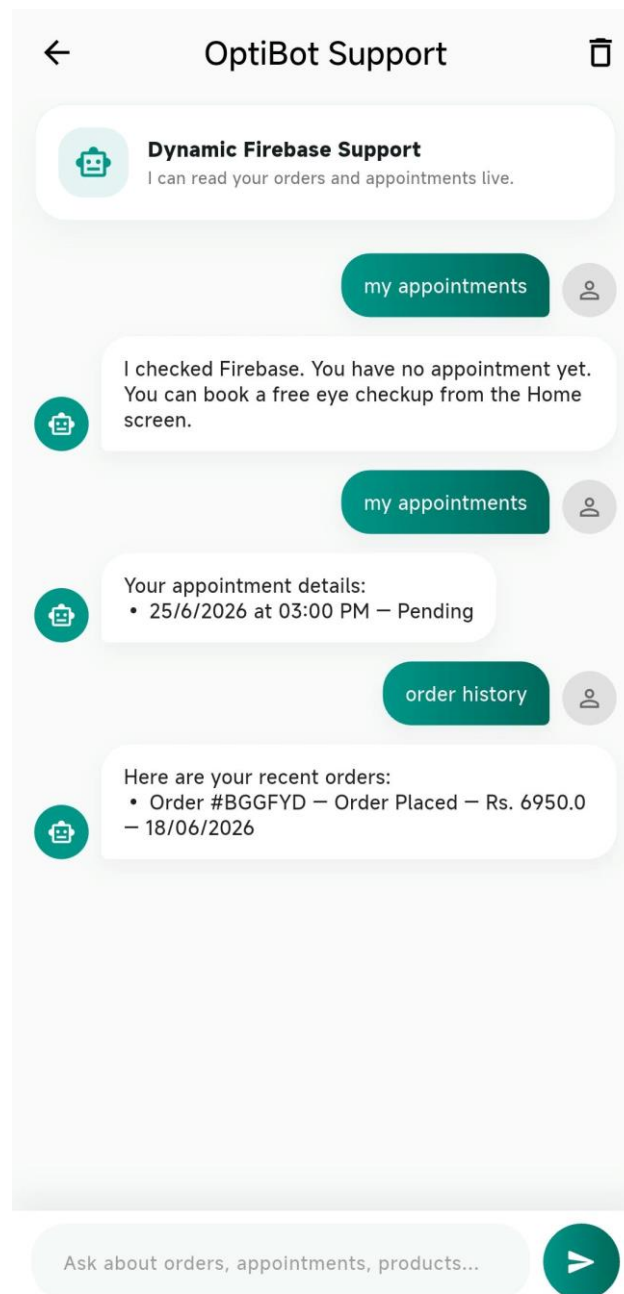


Figure 5.24 UI of OptiBot Support Chatbot Screen

This figure illustrates the "OptiBot Support" chatbot conversational interface within the GAR Opticals application, designed to provide users with automated, live customer assistance. The top header block introduces the chatbot as a "Dynamic Firebase Support" system capable of reading orders and appointments in real time from the backend database. The conversation stream displays alternating message bubbles between the user and the bot, demonstrating natural language processing capabilities to pull appointment details, scheduling updates, and recent order history directly from Firebase collections. At the bottom of the interface, a text input field containing a descriptive placeholder alongside a teal send icon allows users to easily compose and submit custom queries regarding products or services.

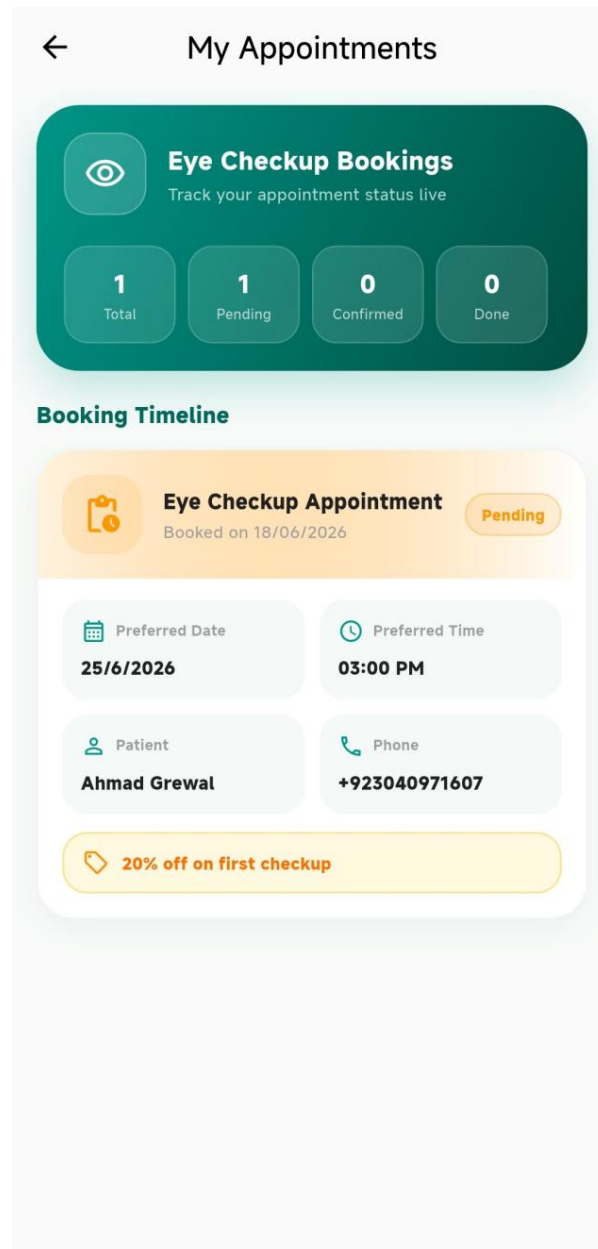


Figure 5.25 UI of My Appointments Screen – Detailed View

This figure displays an analytical view of the "My Appointments" tracker interface within the GAR Opticals application, captured in the reference file image_fe5842.png. The dashboard UI utilizes an organized card-based configuration to separate status tallies from granular appointment details. At the top, an executive summary card displays live metric counters—Total, Pending, Confirmed, and Done—synced with the application's Firebase database structure. The lower timeline segment presents a detailed overview card of a specific scheduling entry, featuring distinct graphical components for the user's selected date, time slot, personal patient details, and contact number, alongside an active promotional tag highlighting applied booking discounts.

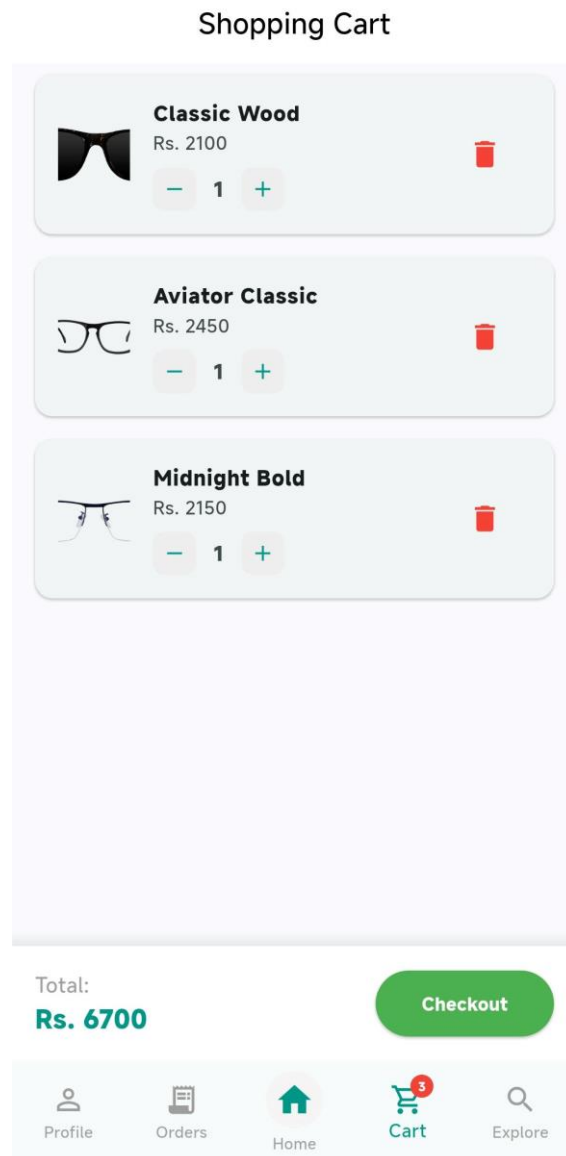


Figure 5.26 UI of Shopping Cart Screen

This figure presents the "Shopping Cart" screen within the GAR Opticals application. The interface displays a vertical scrollable list of selected eyewear products, including "Classic Wood", "Aviator Classic", and "Midnight Bold", with each item enclosed in a distinct card featuring a product thumbnail, pricing metrics, and a red trash bin icon for item removal. Each item row contains interactive decrement and increment toggle buttons to alter purchase quantities, which programmatically updates state variables and recalculates transactional math. The bottom section of the screen includes a sticky summary panel presenting a dynamically computed total cost value alongside a prominent "Checkout" action button, while the bottom navigation bar highlights the active cart view state with an overlay badge indicating item quantity.

←

Checkout

Shipping Information



Full Name



Shipping Address



Phone Number

Payment Method



Cash on Delivery

▼

Order Summary

Subtotal

Rs. 6700

Shipping Fee

Rs. 250

Total Amount

Rs. 6950

Confirm & Place Order

Figure 5.27 UI of Checkout Screen

This figure demonstrates the "Checkout" screen within the GAR Opticals application, as displayed in the reference file image_fdfec0.png. The user interface is structured into three clear data entry and summary blocks to guide the transactional flow. The "Shipping Information" section contains interactive text input fields with prefix icons for capturing the customer's full name, shipping address, and phone number. Below it, the "Payment Method" dropdown component provides transactional flexibility, defaulting to "Cash on Delivery". Finally, the "Order Summary" card dynamically computes and renders the breakdown of financials, totaling the cart subtotal and shipping fees to output a final payable amount, which is processed by the backend upon clicking the prominent teal "Confirm & Place Order" action button.

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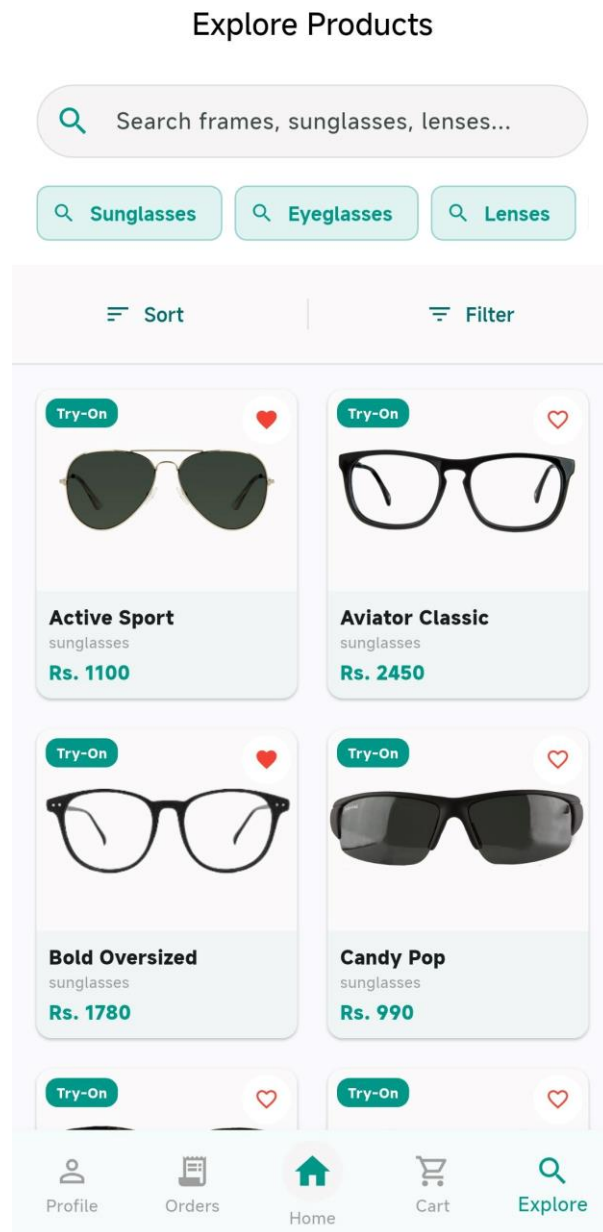


Figure 5.28 UI of Explore Products Screen

This figure presents the "Explore Products" discover screen within the GAR Opticals application. The layout contains a prominent search bar at the top, coupled with horizontal category pills for quick filtering by "Sunglasses", "Eyeglasses", and "Lenses". Below these, interactive "Sort" and "Filter" utility buttons allow for finer product organization. The remaining viewport utilizes a clean two-column grid layout to render active item cards fetched from Firestore—such as "Active Sport", "Aviator Classic", "Bold Oversized", and "Candy Pop"—with each card featuring an overlay "Try-On" badge, product pricing, and interactive favorite heart icons to manage user wishlists.

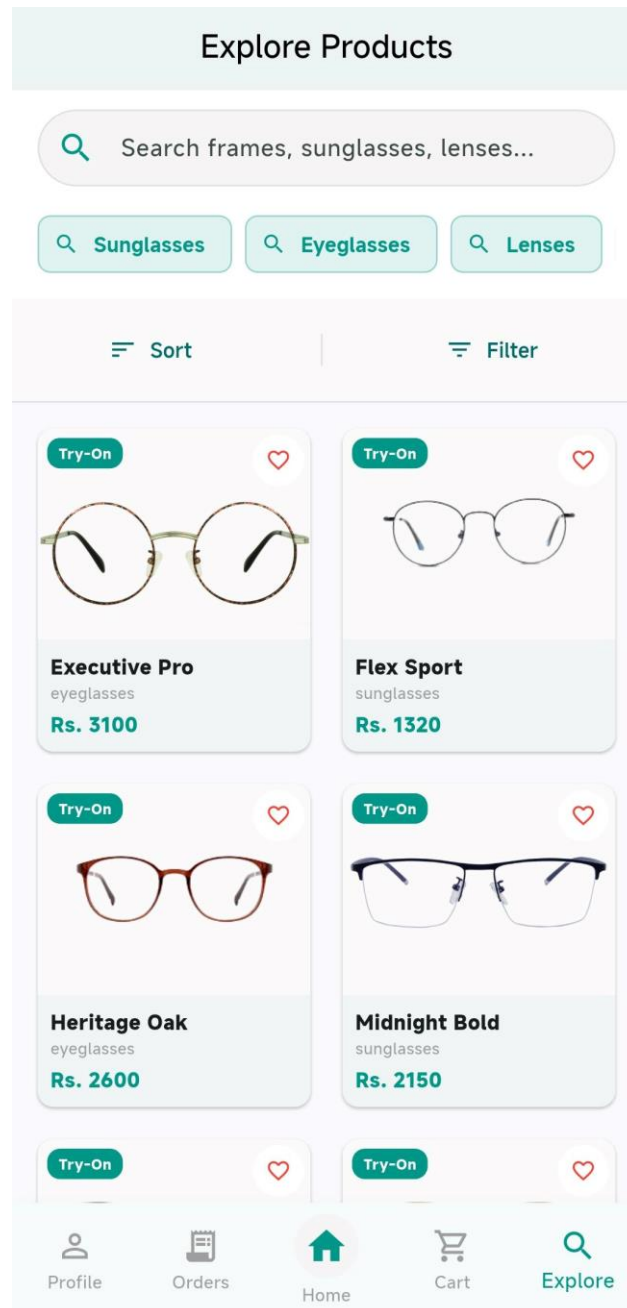


Figure 5.29 UI of Explore Products Screen – Catalog View

This figure presents the "Explore Products" screen within the GAR Opticals application, displaying a comprehensive product catalog layout. The interface includes a top-mounted search field for matching queries against frames, sunglasses, and lenses, sitting directly above horizontal filter chips for specific sub-categories. A secondary row provides quick tap areas for "Sort" and "Filter" modifications. The main content relies on a dynamic grid format pulling directly from the database to present optical items like "Executive Pro", "Flex Sport", "Heritage Oak", and "Midnight Bold", complete with active price displays, responsive wishlist icons, and localized "Try-On" badges for virtual frame placement.

CHAPTER 6

6. Testing and Evaluation

This chapter documents the testing procedures performed to ensure the system's reliability, accuracy, and functional integrity. Testing has been conducted both manually and automatically to verify that the system meets the functional and non-functional requirements established in earlier chapters.

6.1 Manual Testing

6.1.1 System Testing

System testing was performed to verify the end-to-end functionality of the GAR OPTICALS application. The objective was to confirm that all integrated modules (AR, Chatbot, Payment, and Database) communicate correctly. We employed black-box testing techniques to validate user flows from product discovery through to successful order completion.

6.1.2 Unit Testing

Unit testing focused on the smallest testable parts of the application to ensure that individual functions, such as form inputs and basic navigation, work in isolation.

Unit Testing 1: Product Selection Testing Objective: To ensure the product detail page loads correctly when a frame is selected.

Table 6.1 Product Selection Testing Objective

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Verify product page navigation after clicking on a frame image.	Product ID: P001	Product details and description page loads within 2s.	Pass
2.	Verify price calculation on product page.	Quantity: 2	Total price updates to reflect quantity * unit price.	Pass

Unit Testing 2: AR Activation Testing Objective:

To ensure the Virtual Try-On feature initializes on supported devices.

Table 6.2 AR Activation Testing Objective:

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Click on "Try-On" button in product page.	Device: ARCore Compatible	Camera stream initiates and frame overlays correctly.	Pass
2.	Verify permission handling for camera.	Access: Deny	App displays a friendly error message asking for permission.	Pass

6.1.3 Functional Testing

Functional testing was conducted to verify that each module performs the business operations defined in the requirements specification.

Functional Testing 1: User Registration Testing Objective: To ensure users can create accounts and that data is saved to Firebase.

Table 6.3 User Registration Testing Objective

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Register new user via "Sign Up" form.	Email: test@gar.com Password: 123	Firebase creates entry; user redirected to Home Screen.	Pass

Functional Testing 2: Cart Management Testing Objective: To ensure products can be added, viewed, and removed from the cart.

Table 6.4 Cart Management Testing Objective

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Add frame to cart from product page.	Product: Aviator	Item appears in the Cart view.	Pass

6.1.4 Integration Testing

Integration testing ensured that different modules specifically the Payment Gateway and the Firebase Database work together seamlessly.

Integration Testing 1: Checkout and Payment Testing Objective: To ensure transaction data is correctly synced with the backend.

Table 6.5 Checkout and Payment Testing Objective

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Proceed to checkout and complete payment.	Method: Stripe / Test Card	Payment successful; Order record created in Firestore.	Pass
2.	Check database for order update.	Order ID: ORD123	Status in Firestore updates to "Pending"	Pass
3.	View order tracking history.	User: Current User	Order history displays the current status accurately.	Pass

6.2 Automated Testing

Automated testing was utilized to verify code logic and UI stability across multiple screen sizes to reduce manual regression time.

Table 6.6 Summary of Automated Testing Tools Used

Tool Name	Tool Description	Applied on [list of related test cases / FR / NFR]	Results
Flutter Test	Unit testing framework for Dart code	FR-1 (Auth logic), FR-2 (Catalog logic)	All tests passed
Firebase Test Lab	Cloud-based mobile app testing	NFR-Performance, NFR-Compatibility	No crashes reported
Integration Test	Automated UI flow testing	FR-4 (Order placement flow)	Successful end-to-end

CHAPTER 7

7. Conclusion and Future Work

7.1 Conclusion

The GAR OPTICALS system has successfully achieved its primary objective of modernizing the traditional optical retail experience through digital transformation. By integrating Augmented Reality (AR) and Artificial Intelligence (AI) into a user-friendly cross-platform mobile application, the project has effectively addressed the core challenges of visualization uncertainty and operational inefficiency.

The implementation of the AR-based Virtual Try-On feature has provided customers with a reliable, real-time visualization tool that bridges the gap between physical and digital shopping. Furthermore, the integration of an AI-driven recommendation engine and an automated chatbot has significantly enhanced the personalization and responsiveness of the customer support experience. The backend infrastructure, powered by Firebase, has proven effective in ensuring real-time data synchronization, secure user authentication, and scalable management for store administrators via the web-based dashboard.

In summary, GAR OPTICALS serves as a benchmark for small-to-medium optical enterprises, demonstrating that advanced technologies like AR and AI can be effectively leveraged to optimize retail processes, improve customer engagement, and streamline inventory and order management.

7.2 Future Work

While the current version of the GAR OPTICALS system meets the defined requirements, the architecture is designed to be modular, allowing for several enhancements that could be integrated in future development cycles:

- **Integration with Courier APIs:** Future versions can integrate with local or international courier service APIs to automate shipping labels, real-time logistics, and last-mile delivery tracking directly within the app.
- **Advanced Facial Measurement:** Incorporating deeper computer vision algorithms to provide precise PD (Pupillary Distance) and face-shape measurements would further enhance the accuracy of the AR try-on feature and prescription lenses fitting.
- **Multi-Vendor Expansion:** The system could be scaled to support a multi-vendor marketplace, allowing multiple independent optical stores to join the platform and manage their own inventory.

- **Multilingual Support:** To reach a broader audience, the application can be updated to support multiple local and international languages, particularly to accommodate diverse user bases.
- **Offline Functionality Expansion:** Future iterations could implement more robust local caching of 3D AR assets, allowing users to perform virtual try-ons even in scenarios with intermittent or poor internet connectivity.
- **Integration of Social Commerce Features:** Future updates will aim to include "Social Sharing" capabilities, allowing users to share their AR try-on photos directly to platforms like Instagram and WhatsApp. This not only increases user engagement but also serves as an organic marketing tool for the brand, driving new traffic to the application through peer recommendations.
- **AI-Based Prescription Scanning:** To further simplify the user journey, a future iteration could include an OCR (Optical Character Recognition) module. This would allow users to simply upload a photo of their physical prescription, which the system would then automatically scan and parse into the digital order form, reducing manual entry errors and improving order accuracy.
- **Virtual Eye Testing and PD Measurement:** Beyond visualization, the application can be expanded to include basic vision screening tools and automated Pupillary Distance (PD) measurement using the device's depth-sensing camera.

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